*(Labguide) of instruments and supplies. The most relevant reviews to this unit are Coatings and Surface Characterization reviews in the Application Reviews issue (published odd years) and Ultraviolet and Light Absorption Spectrometry review in the Fundamental Reviews issue (published even years). An on-line version of the Labguide is currently available at http://pubs.acs.org/labguide/.*

Demtröder, W. 1996. Laser Spectroscopy: Basic Concepts and Instrumentation, 2nd ed. Springer-Verlag, Berlin.

*A graduate-level text on laser spectroscopy, including theory, laser-spectroscopic techniques, and instrumentation. Provides an overview of advanced absorption spectroscopies and competitive techniques such as photoacoustic spectroscopy.*

Hollas, J. M. 1996. Modern Spectroscopy, 3rd ed. John Wiley & Sons, New York.

*A graduate-level text that covers the principles and instrumentation of spectroscopy. Specific topics include quantum mechanics and interaction of light and matter, molecular symmetry, rotational spectroscopy, electronic spectroscopy, vibrational spectroscopy, photoelectron and surface spectroscopies, and laser spectroscopy.*

Ingle and Crouch, 1988. See above.

*A graduate-level text that covers the principles and instrumentation of spectroscopy. Specific techniques include atomic absorption and emission spectroscopy, molecular absorption and fluorescence spectroscopy, infrared absorption spectroscopy, and Raman scattering.*

Settle, F. A. 1997. Handbook of Instrumental Techniques for Analytical Chemistry. Prentice-Hall, New York.

*A broad comprehensive handbook of analytical techniques. It covers separation techniques, optical spectroscopy, mass spectrometry, electrochemistry, surface analysis, and polymer analysis.*

## APPENDIX: GLOSSARY OF TERMS AND SYMBOLS USED

| | |
|---|---|
| $a$ | Absorptivity |
| $A$ | Absorbance, $= abc = \varepsilon bc$ |
| A.U. | Absorbance unit |
| $b$ | Path length |
| $c$ | Analyte concentration; also velocity of light in vacuum ($2.99795 \times 10^8$ m/s), $= \lambda\nu$ |
| $D_4$, $D_5$ | Deformation potentials |
| $E$ | Energy of a photon, $= h\nu$ |
| $E_{AB}$ | Energy separation of A and B bands |
| $h$ | Planck's constant ($6.626 \times 10^{-34}$ Js) |
| $I$ | Light intensity |
| $I_0$ | Initial light intensity |
| $k$ | Absorption coefficient |
| $P$ | Radiant power (radiant energy on unit area in unit time) |
| $P_0$ | Initial radiant power; power transmitted through sample holder and solvent only |
| $P_s$ | Power of source light |
| $r_\infty$ | Relative reflectance, $= r_{sample}/r_{reference}$ |
| $R$ | Reflectance (includes diffuse and specular reflectance) |
| $\%R$ | Percent reflectance |
| $s$ | Scattering coefficient |
| $T$ | Transmittance |
| $\%T$ | Percent transmittance |
| $\varepsilon$ | Molar absorptivity |
| $\lambda$ | Wavelength |
| $\nu$ | Frequency |

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# RAMAN SPECTROSCOPY OF SOLIDS

## INTRODUCTION

Raman spectroscopy is based on the inelastic scattering of light by matter and is capable of probing the structure of gases, liquids, and solids, both amorphous and crystalline. In addition to its applicability to all states of matter, Raman spectroscopy has a number of other advantages. It can be used to analyze tiny quantities of material (e.g., particles that are $\sim$1 μm on edge), as well as samples exposed to a variety of conditions such as high temperature and high pressure and samples embedded in other phases, so long as the surrounding media are optically transparent.

Succinctly stated, Raman scattering results from incident radiation inducing transitions in the atoms/molecules that make up the scattering medium. The transition can be rotational, vibrational, electronic, or a combination (but first-order Raman scattering involves only a single incident photon). In most studies of solids by Raman spectroscopy, the transitions observed are vibrational and these will be the focus of this unit.

In a Raman experiment, the sample is irradiated with monochromatic radiation. If the sample is transparent, most of the light is transmitted, a small fraction is elastically (Rayleigh) scattered, and a very small fraction is inelastically (Raman) scattered. The inelastically scattered light is collected and dispersed, and the results are presented as a Raman spectrum, which plots the intensity of the inelastically scattered light as a function of the shift in wavenumber of the radiation. (The wavenumber of a wave is the reciprocal of its wavelength and is proportional to its momentum in units of reciprocal centimeters.) Each peak in the spectrum corresponds to one or more vibrational modes of the solid. The total number of peaks in the Raman spectrum is related to the number of symmetry-allowed, Raman active modes. Some of the modes may be degenerate and some may have Raman intensities that are too low to be measured, in spite of their symmetry-allowed nature. Consequently, the number of peaks in the Raman spectrum will be less than or equal to the number of Raman active modes. The practical usefulness of Raman spectroscopy resides largely in the fact that the Raman spectrum serves as a fingerprint of the scattering material. In fact, Raman activity is a function of the point group symmetry of a molecule and the space group symmetry of a crystalline solid; it can provide a range of information, including the strength of interatomic and intermolecular bonds, the mechanical strain present in a solid, the

composition of multicomponent matter, the degree of crystallinity of a solid, and the effects of pressure and temperature on phase transformations.

### Competitive and Related Techniques

As mentioned above, Raman spectroscopy is extremely versatile and can be used to investigate the structures of solids, liquids, and gases. This unit is concerned with the analyses of solids. In general, two major types of experimental techniques are used to obtain information about the structures of solids: spectroscopy and diffraction. Spectroscopy is based on the interaction of electromagnetic radiation with matter. Diffraction occurs whenever a wave is incident on an array of regularly spaced scatterers in which the wavelength of the wave is similar to the spacing of the scatterers.

X rays, electrons, and neutrons with wavelengths similar to the interatomic spacing of a crystal will be diffracted by the crystal. The diffraction pattern will provide detailed information about, e.g., the crystal structure, crystal size, interplanar spacings, long- and short-range order, and residual lattice strain. While diffraction can provide some information about gases, liquids, and amorphous solids, the greatest amount of information is obtained for crystals.

Spectroscopic methods can probe the electronic, vibrational, rotational, and nuclear states of matter. In one family of spectroscopic techniques, the material absorbs electromagnetic radiation of energy equivalent to the difference in energy between particular states of the material. For example, transitions between adjacent electronic states generally require energies equivalent to that of ultraviolet and visible radiation (UV-vis electronic spectroscopy; see ULTRAVIOLET AND VISIBLE ABSORPTION SPECTROSCOPY). Vibrational transitions typically make use of infrared radiation (infrared absorption spectroscopy, IRAS). In the gas-phase molecules may absorb microwave radiation and undergo a change in rotational state. Finally, in nuclear magnetic resonance (NMR) the nuclei of atoms with angular momentum (spin) and magnetic moment absorb energy in the radio frequency region while subjected to magnetic fields that are alternating synchronous with the natural frequencies of the nucleus (see NUCLEAR MAGNETIC RESONANCE IMAGING).

Raman spectroscopy, unlike the above-mentioned spectroscopies, involves the scattering of electromagnetic radiation rather than absorption. In classical terms, the energy of the Raman scattered radiation is shifted from that of the incident radiation because of modulation by the vibrations of the scattering medium. Thus, Raman spectroscopy (RS), like IRAS, probes the vibrational spectra of materials. Because IRAS involves absorption of radiation and RS involves inelastic scattering of radiation, the two techniques are complementary. Since Raman spectroscopy makes use of optical radiation, it is a highly flexible technique and can probe the structures of materials in relatively inaccessible locations, such as in high-pressure chambers, high-temperature furnaces, and even aqueous solutions. All that is necessary is the ability to get laser light onto the sample and to collect the scattered light.

For materials that are both Raman active and IR active, the Raman scattering cross-section is much smaller than the IR absorption cross-section so the incident radiation needs to be more intense and the detectors need to be more sensitive for Raman spectroscopy than for IR spectroscopy. Because of the fundamental differences between absorption and scattering of electromagnetic radiation, some vibrational modes of a material may be Raman active but not IR active. In particular, for a molecule with a center of symmetry, vibrational modes that are Raman active are IR inactive and vice versa. Consequently, when deciding between IR and Raman spectroscopies, the point group or space group symmetry of the material needs to be taken into account.

A major advantage of Raman spectroscopy compared to x-ray diffraction is the ability of Raman spectroscopy to provide detailed structural information of amorphous materials. The Raman spectrum of an amorphous solid, for example, is directly related to its complete density of (vibrational) states. However, all metals are Raman inactive so Raman spectroscopy cannot be used for the structural analyses of this entire class of materials.

## PRINCIPLES OF THE METHOD

### Theoretical Background

This discussion of Raman spectroscopy will begin with a description of the fundamental phenomenon that underlies the scattering of light by matter, namely the generation of electromagnetic radiation by the acceleration of charged particles. As a light-scattering process, Raman scattering is rigorously described by treating both the incident and scattered radiation and the scattering medium quantum mechanically. In this unit a full quantum mechanical description of Raman scattering is not given. Instead, Raman scattering is treated less rigorously but in a manner that allows all of the practically important phenomena associated with Raman spectroscopy of solids to be identified. Specifically, a combination of classical physics and semiclassical physics (treating the scattering medium quantum mechanically while approximating light as a sinusoidally varying electric field) is used to describe Raman scattering.

**Generation of Electromagnetic Radiation—Classical Physics.** Classical physics dictates that accelerating charged particles emit electromagnetic radiation whose electric field $\varepsilon$ (a vector quantity) is directly proportional to the particle's acceleration $\mathbf{a}$ (a vector quantity; see Fig. 1; Rossi, 1957):

$$\varepsilon_{\mathbf{r}}(r,\theta,\phi,t) = \varepsilon_{\phi}(r,\theta,\phi,t) = 0 \tag{1a}$$

and

$$\varepsilon_{\theta}(r,\theta,\phi,t) = \left(\frac{q\sin\theta}{4\pi c^2 r}\right)a \tag{1b}$$

where $c$ is the velocity of light; $a$ is the magnitude of the acceleration of the charged particle of charge $q$; $r$, $\theta$, and $\phi$ are the spherical coordinates with the particle located at

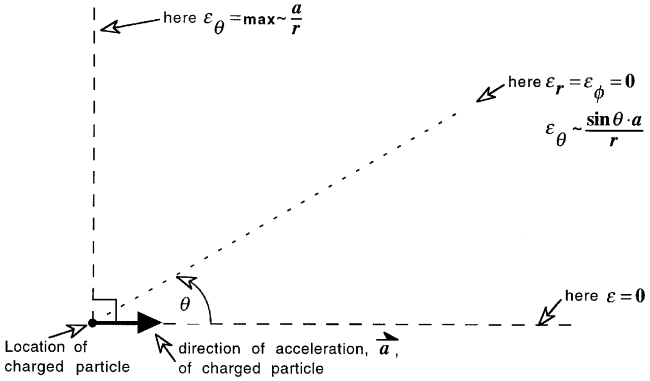



**Figure 1.** Electromagnetic radiation associated with an accelerating charged particle.

the origin and; $t$ is time. The term $\varepsilon_i(r, \theta, \phi, t)$ is the $i$th component of the electric field at the point ($r$, $\theta$, $\phi$) and time $t$. The other terms are defined in Figure 1.

The optical behavior of matter, e.g., the emission of sharp spectral lines by elemental gases excited by an electric discharge, requires that matter, when treated by classical physics, be composed of charged particles that behave as harmonic oscillators when displaced from their equilibrium positions. When matter is illuminated by monochromatic optical radiation, the electrons are driven from their equilibrium positions by the sinusoidally varying electric field of light, creating a time-varying electric polarization $\mathbf{p}$ (a vector quantity) that is directly proportional to the acceleration of the electrons. Approximating the polarized molecule by a dipole, $\mathbf{p}$ is given by

$$\mathbf{p} = -e\mathbf{x} = -e\mathbf{x}_0 \cos \omega_{\text{inc}} t \tag{2a}$$

where $\mathbf{x}$ is the displacement of the center of negative charge of the molecule from the center of positive charge and $\mathbf{x}_0$ is the maximum value of this displacement.

$$\mathbf{p} = -\left(\frac{\mathbf{e}}{\omega_{\text{inc}}^2}\right)\mathbf{a} \tag{2b}$$

where $-e$ is the electric charge of the electron and $\omega_{\text{inc}}$ is the frequency of oscillation of the electron, which matches the frequency of the incident light. The electrons thereby emit radiation whose electric field is proportional to the polarization of the molecule:

$$\varepsilon_\theta \left(\frac{\omega_{\text{inc}}^2 \sin\theta}{4\pi\varepsilon_0 c^2 r}\right)|\mathbf{p}| \tag{3}$$

where $\mathbf{x}$ is the displacement of the center of negative charge of the molecule from the center of positive charge and $\mathbf{x}_0$ is the maximum value of this displacement.

$$S_{\text{r}} = \left(\frac{\omega_{\text{inc}}^4 \sin^2\theta}{16\pi^2\varepsilon_0^2 c^4 r^2}\right)|\mathbf{p}|^2 \tag{4}$$

where $\varepsilon_0$ is the dielectric strength of the medium in which the radiation is propagating and $S_{\text{r}}$ is the radial component of the Poynting vector, which is the only nonzero component of the Poynting vector. The Poynting vector is the energy flux of the light wave. It is equal to the amount of energy flowing through a unit area in unit time.

When analyzing the effects of irradiating a molecule or a large quantity of matter with electromagnetic radiation, it is not necessary to treat each electron individually. Rather, it is possible to model the molecule(s) by atoms that are composed of heavy, positively charged ion cores and a mobile negative charge, which together form an electric dipole. If the centers of positive and negative charges are not coincident, the molecule has a permanent dipole moment. If the negative charge is displaced from its equilibrium position, e.g., by colliding with another particle, it will harmonically oscillate about its equilibrium position and emit electromagnetic radiation until all of the energy it gained in the collision is radiated away. Similarly, if the molecule is irradiated with an electromagnetic wave, the negative charge will be driven away from its equilibrium position by the applied electric field and will harmonically oscillate about its equilibrium position. The oscillating electric dipole will emit electromagnetic radiation with an electric field given by Equation 4 with $|\mathbf{p}| = p_0 \cos \omega_{\text{inc}} t$, where $\omega_{\text{inc}}$ is the frequency of the incident radiation and $p_0$ is the magnitude of the polarization when the centers of positive and negative charges experience their maximum separation.

The polarization $\mathbf{p}$ induced by the applied electric field of the incident electromagnetic radiation varies with time at the same frequency as the applied electric field. There is a second contribution to the time dependency of the material's polarization, namely that caused by the time-dependent nuclear displacements (thermally excited molecular vibrations), which, as illustrated in Figure 2, act to modulate the radiation generated by the electrons oscillating under the influence of the incident radiation (Long, 1977). As a result, there are three frequency components to the scattered light, which is the radiation emitted by the time-varying, molecular electric dipoles. One component (the Rayleigh line) has the same frequency, $\omega_{\text{inc}}$ as the incident radiation and the other two components have a slightly lower and a slightly higher frequency, $\omega_{\text{inc}} + \omega_{\text{v}}$ and $\omega_{\text{inc}} - \omega_{\text{v}}$, respectively, where $\omega_{\text{v}}$ is the vibrational frequency of the atoms/molecules making up the scattering medium. These last two components represent the Stokes and anti-Stokes components, respectively, of the Raman scattered radiation.

Classically, the expression for the intensity of the Raman scattered radiation is

$$I = \langle |S| \rangle \frac{dA}{d\Omega} \tag{5}$$

where $A$ is the surface area of the sphere centered at the source of the Raman scattered radiation, $\Omega$ is the solid angle through which the radiation passes, and $\langle |S| \rangle$ is the time-averaged value of the Poynting vector of the Raman scattered radiation. Its form is identical to that given above for $S_{\text{r}}$ except that the time-varying part of $\mathbf{p}$ (see Equation 4) is now given by

$$\mathbf{p} \sim \left(\frac{\partial\alpha}{\partial\mathbf{Q}_k}\right)_0 \mathbf{Q}_k \boldsymbol{\varepsilon} \tag{6}$$

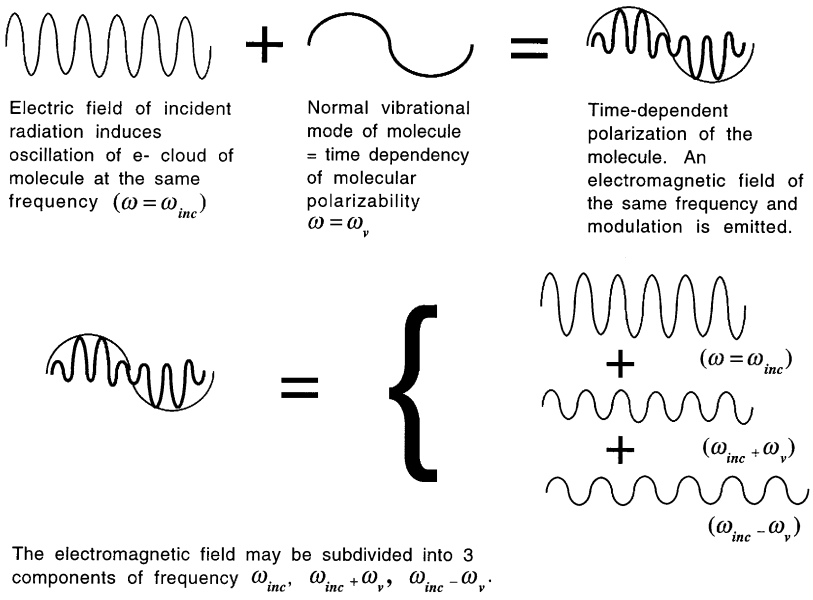



**Figure 2.** Polarization of vibrating molecule irradiated by light.

where $\underline{\alpha}$ (a tensor quantity) is the polarizability of the material, $\mathbf{Q}_k$ (a vector quantity) is the normal coordinate of the $k$th vibrational mode, and $(\partial\underline{\alpha}/\partial\mathbf{Q}_k)_0$ (a tensor quantity) is the differential polarizability, which indicates the substance's change in polarizability at its equilibrium position during the $k$th normal mode of vibration. Consequently, the material property that is directly probed by Raman spectroscopy is the differential polarizability $(\partial\underline{\alpha}/\partial\mathbf{Q}_k)_0$. It is a symmetric tensor quantity, and at least one component must be nonzero for Raman scattering to occur. Knowing the point group symmetry of a molecule or the space group symmetry of a crystalline solid, it is possible to use group theoretical techniques to determine if the symmetry permits nonzero components of the differential polarizability tensor. The use of molecular and crystal symmetry greatly simplifies the analyses of Raman scattering by a material, as will be demonstrated below (see Overview of Group Theoretical Analysis of Vibrational Raman Spectroscopy).

**Light Scattering—Semiclassical Physics.** From the perspective of semiclassical physics, Raman scattering results from the incident radiation inducing a transition (considered here to be vibrational) in the scattering entity, as schematically illustrated in Figure 3. The material is placed by the sinusoidally varying electric field into a higher energy state, termed a virtual state, which may be thought of as the distortion of the molecule by the electric field. As shown in Figure 3, when the molecule returns to a lower energy state, it emits radiation of frequency $\omega_{\text{inc}}$(=> Rayleigh, or elastic scattering), $\omega_{\text{inc}} - \omega_v$(=> Stokes Raman scattering), or $\omega_{\text{inc}} + \omega_v$ (=> anti-Stokes Raman scattering). The probability of these transitions occurring is, according to quantum mechanical perturbation theory, given by $a_{mn}^2$, where

$$a_{mn} \sim \int \psi_m^* \mathbf{p} \psi_n dV \tag{7}$$

Here, $\psi_i$ is the wave function of the $i$th state and $\psi_i^*$ is its complex conjugate.

$$\mathbf{p} = \underline{\alpha}\boldsymbol{\varepsilon} \tag{8}$$

When expressed in terms of the individual components, $p_i$, Equation 8 becomes

$$p_i = \sum \alpha_{ij}\varepsilon_j \tag{9}$$

where $\alpha_{ij}$ is the $ij$ matrix component of the polarizability, $\varepsilon_j$ is the $j$ component of the electric field, and the differential $dV$ includes differential displacements of all quantum vibrational states. For Rayleigh scattering, $m = n$ is the vibrational ground state; for Stokes Raman scattering, $n$ is the vibrational ground state and $m$ is the first vibrational excited state; for anti-Stokes Raman scattering, $n$

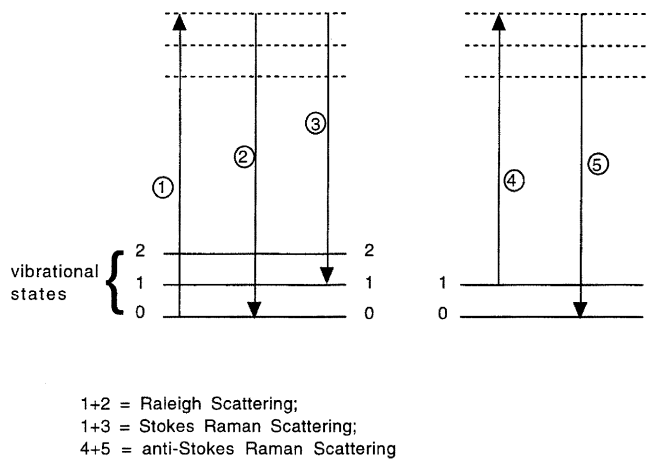



**Figure 3.** Rayleigh and Raman scattering. $1 + 2$, Rayleigh scattering; $1 + 3$, Stokes Raman scattering; $4 + 5$, anti-Stokes Raman scattering.

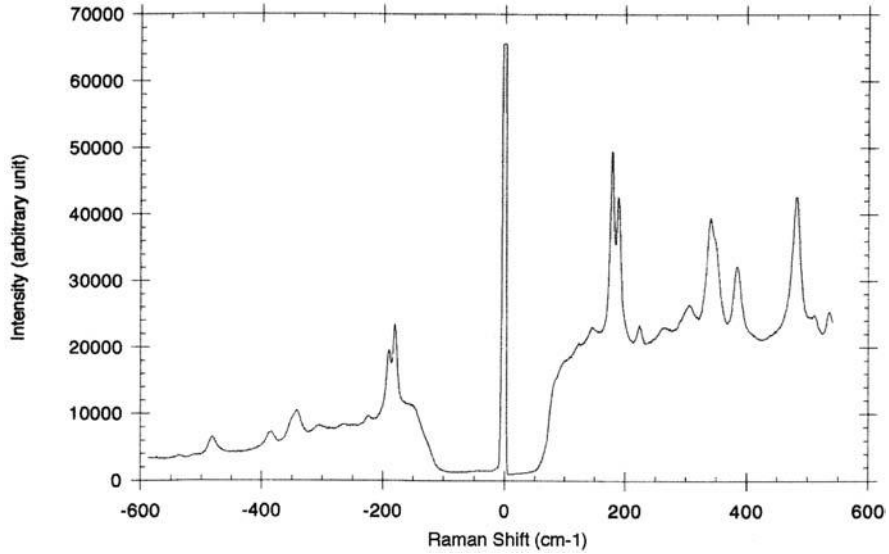



**Figure 4.** Raman spectrum of impure, polyphase (monoclinic+tetragonal) $ZrO_2$. (C. S. Kumai, unpublished results, 1998.)

is the first vibrational excited state and $m$ is the vibrational ground state.

Since the population of energy levels in a collection of atoms/molecules is governed by the Maxwell-Boltzmann distribution function, the population of the vibrational ground state will always be greater than the population of the first vibrational excited state. As a result, the intensity of the Stokes-shifted Raman scattering will always be greater than the intensity of the anti-Stokes-shifted Raman scattered radiation. This effect, which is predicted quantum mechanically but not classically, is illustrated in Figure 4, which presents the Raman spectrum of polycrystalline $ZrO_2$ (mixed monoclinic and tetragonal). The Raman spectrum is depicted in terms of the shift (in wavenumbers) in the wavelength of the scattered radiation with respect to that of the incident light. Thus the Rayleigh peak is seen in the center, and the anti-Stokes and Stokes lines are to the left and right, respectively. Note that for each peak in the Stokes Raman section of the spectrum there is a corresponding peak of lower intensity in the anti-Stokes Raman section.

## Overview of Group Theoretical Analysis of Vibrational Raman Spectroscopy

The use of group theory in determining whether or not a given integral can have a value of zero may be illustrated using the ground vibrational state $\psi_0(x)$ and first vibrational excited state $\psi_1(x)$ of a harmonic oscillator. The equations describing these two vibrational states are, respectively,

$$\psi_0(x) = \exp\left(-\frac{2\pi m\omega x^2}{\hbar}\right) \tag{10}$$

$$\psi_1(x) = \left(\frac{4\pi^{1/2} m\omega}{\hbar}\right)^{1/2} \exp\left(-\frac{2\pi m\omega x^2}{\hbar}\right) \tag{11}$$

where $\hbar = h/2\pi$ and $h$ is Planck's constant. These two vibrational states are plotted in Figure 5.

By inspection,

$$\int_{-\infty}^{\infty} \psi_1(x)dx = 0 \tag{12}$$

That is, performing the integration by summing up the infinite number of terms given by

$$\sum \psi_1(x_n)\Delta x_n \tag{13}$$

(where $\Delta x_n = x_n - x_{n-1}$), it is seen that for every positive contribution (to the sum) given by $\psi_1(x_i)\Delta x_i$ there is a term equal in magnitude but opposite in sign [$\psi_1(x_{-i})\Delta x_{-i} = -\psi_1(x_i)\Delta x_i$].

Thus, the entire sum in Equation 13 is zero.

Similarly, by inspection,

$$\int_{-\infty}^{\infty} \psi_0(x)dx \neq 0 \tag{14}$$

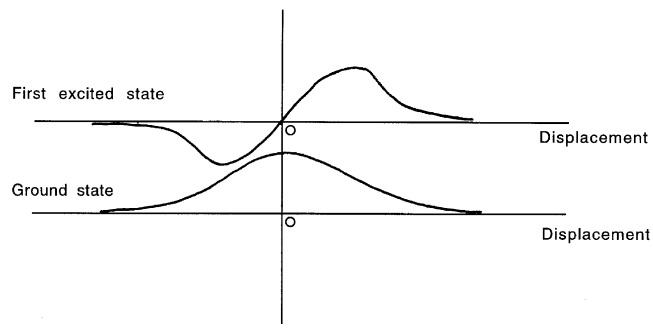



**Figure 5.** Ground-state vibrational wave function and first excited-state vibrational wave function of harmonic oscillator.

That is, every term $\psi_0(x_i)\Delta x_i$ in the sum $\sum \psi_0(x_n)\Delta x_n$ has the identical sign and, hence,

$$\int_{-\infty}^{\infty} \psi_0(x)dx \approx \sum \psi_0(x_n)\Delta x_n \neq 0 \qquad (15)$$

The above analysis of the integrals in Equations 12 and 13 can be repeated using group theory. To begin, there needs to be a mathematical description of what is meant by the symmetry of a function. The difference between the symmetry of $\psi_1(x)$ and $\psi_0(x)$ can be expressed as follows. Imagine that there is a mirror plane parallel to the *YZ* plane and that passes through the origin, (0, 0, 0); then the operation of this mirror plane on the function $\psi_0(x)$ is described by the phrase "a mirror plane acting on $\psi_0(x)$ yields $\psi_0(x)$."

If the effect of the mirror plane acting on $\psi_0(x)$ is represented by a matrix $[R]$, then

$$\text{Effect of mirror plane acting on } \psi_0(x) = [R][\psi_0(x)] \qquad (16)$$

In this case, $[R] = [I]$ is the identity matrix.

Similarly, for $\psi_1(x)$, the effect of a mirror plane acting on $\psi_1(x)$ equals $-\psi_1(x)$. In matrix notation, this statement is given as

$$\text{Effect of mirror plane acting on } \psi_1(x) = [R]\psi_1(x) = -[I]\psi_1(x) \qquad (17)$$

Note that the effect of the mirror operation on $x$ is to produce $-x$ and that $\psi(-x) = -\psi(x)$. The character of the one-dimensional matrix in Equation 16 is $+1$ and that in Equation 17 is $-1$. Consider now a molecule that exhibits a number of symmetry elements (e.g., mirror planes, axes of rotation, center of inversion). If the operation of each of those on a function leaves the function unchanged, then the function is said to be "totally symmetric" with respect to the symmetry elements of the molecule (or, in other words, the function is totally symmetric with respect to the molecular point group, which is the collection of all the molecule symmetry elements). For the totally symmetric case, the character of each one-dimensional (1D) matrix representing the effect of the operation of each symmetry element on the function is 1 for all symmetry operations of the molecule. For such a function, the value of the integral over all space of $\psi(x)dx$ will be nonzero.

If the operation of any symmetry element of the molecule (in the molecular point group) generates a matrix that has a character other than 1, the function is not totally symmetric and the integral of $\psi(x)dx$ over all space will be zero.

Now consider the application of group theory to vibrational spectroscopy. A vibrational Raman transition generally consists of the incident light causing a transition from the vibrational ground state to a vibrational excited state. The transition may occur if

$$\int_{-x}^{x} \psi_1(x)\alpha_{ij}\psi_0(x)dx \neq 0 \qquad (18)$$

**Table 1. Character Table of Point Group $C_{2v}$**

| $C_{2v}$ | $E$ | $C_2(z)$ | $\sigma_v(xz)$ | $\sigma'_v\ (yz)$ | Basis Functions |
|---|---|---|---|---|---|
| $A_1$ | $+1$ | $+1$ | $+1$ | $+1$ | $z; x^2; y^2; z^2$ |
| $A_2$ | $+1$ | $+1$ | $-1$ | $-1$ | $xy$ |
| $B_1$ | $+1$ | $-1$ | $+1$ | $-1$ | $x; xz$ |
| $B_2$ | $+1$ | $-1$ | $-1$ | $+1$ | $y; yz$ |

That is, the integral may have a nonzero value if the integrand product is totally symmetric over the range of integration. The symmetry of a function that is the product of two functions is totally symmetric if the symmetries of the two functions are the same. In addition, for the sake of completeness, the last statement will be expanded using group theoretical terminology, which will be defined below (see Example: Raman Active Vibrational Modes of $\alpha$-$Al_2O_3$). The symmetry species of a function that is the product of two functions will contain the totally symmetric irreducible representation if the symmetry species of one function contains a component of the symmetry species of the other.

The ground vibrational state is totally symmetric. Hence, the integrand is totally symmetric and the vibrational mode is Raman active if the symmetry of the excited vibrational state $\psi_1(x)$ is the same as, or contains, the symmetry of the polarizability operator $\alpha_{ij}$.

As shown below (see Appendix), the symmetry of the operator $\alpha_{xy}$ is the same as that of the product $xy$. Thus, the integrand $\psi_1\alpha_{xy}\psi_0$ is totally symmetric if $\psi_1$ has the same symmetry as $xy$ (recall that $\psi_0$, the vibrational ground state, is totally symmetric).

A molecular point group is a mathematical group (see Appendix) whose members consist of all the symmetry elements of the molecule. Character tables summarize much of the symmetry information about a molecular point group. Complete sets of character tables may be found in dedicated texts on group theory (e.g., Bishop, 1973). The character table for the point group $C_{2v}$ is presented in Table 1. In the far right-hand column are listed functions of interest in quantum mechanics and vibrational spectroscopy in particular. The top row lists the different symmetry elements contained in the point group and illustrated in Figure 6. Here, $C_2$ is a twofold rotational axis that is parallel to the $z$ axis. The parameters $\sigma_{xz}$ and $\sigma_{yz}$ are mirror planes parallel to the $xz$ and $yz$ planes, respectively. In each row beneath the top row are listed a set of numbers. Each number is the character of the matrix that represents the effect of the symmetry operation acting on any of the functions listed in the right-hand end of the same row. In the point group $C_{2v}$, e.g., the functions $z$, $x^2$, $y^2$, and $z^2$ have the same symmetry, which in the terminology of group theory is identified as $A_1$. The functions $y$ and $yz$ have the same symmetry, $B_2$.

As shown in Figure 6, the water molecule, $H_2O$, exhibits the point group symmetry $C_{2v}$. There are three atoms in $H_2O$ and hence there are $3N - 6 = 3$ normal vibrational modes, which are depicted in Figure 7. These exhibit symmetries $A_1$, $A_1$, and $B_2$. That is, the vibrational wave functions of the first excited state of each of these modes

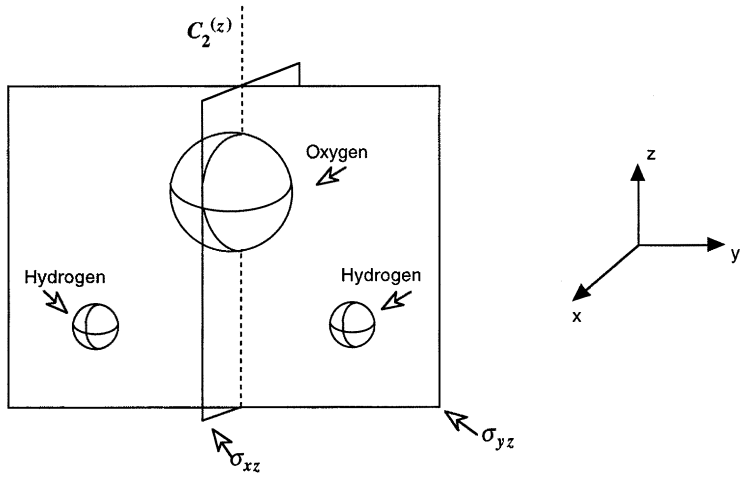



**Figure 6.** Symmetry operations of $H_2O$ (point group $C_{2v}$).

possess the symmetries $A_1$, $A_1$, and $B_2$, respectively. Since $x^2$, $y^2$, and $z^2$ exhibit $A_1$ symmetry, so too would $\alpha_{xx}$, $\alpha_{yy}$ and $\alpha_{zz}$. Hence the product of $\alpha_{xx}$, $\alpha_{yy}$ and $\alpha_{zz}$ with $\psi_1$ for the vibrational modes with $A_1$ symmetry would be totally symmetric, meaning that the integral over the whole space $\psi_{A1}\alpha_{xx(\text{or}yy,zz)}\psi_0 dV$ may be nonzero, meaning that this vibrational mode is Raman active (the integrand $\psi_{A1}\alpha_{xx(\text{or}yy,zz)}\psi_0 dV$ is the same as the integrand in Equation 7).

Now consider the Raman vibrational spectra of solids. One mole of a solid will have $3 \times 6.023 \times 10^{23} - 6$ normal modes of vibration. Fortunately, it is not necessary to consider all of these. As demonstrated below (see Appendix), the only Raman active modes will be those near the center of the Brillouin zone (BZ; $k = 0$). Solids with only one atom per unit cell have zero modes at the center of the BZ and hence are Raman inactive (see Appendix). Solids with more than one atom per unit cell, e.g., silicon and $Al_2O_3$, are Raman active.

The symmetry of a crystal is described by its space group. However, as demonstrated below (see Appendix), for the purposes of vibrational spectroscopy, the symmetry of the crystal is embodied in its crystallographic point group. This is an extremely important and useful conclusion. As a consequence, the above recipe for deciding the Raman activity of the vibrational modes of $H_2O$ can be applied to the vibrational modes of any solid whose unit cell is multiatomic.

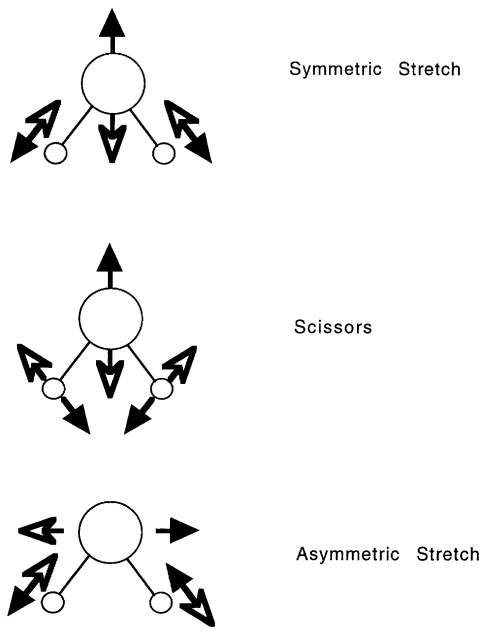



**Figure 7.** Normal vibrational modes of $H_2O$.

## PRACTICAL ASPECTS OF THE METHOD

The weakness of Raman scattering is the characteristic that most strongly influences the equipment and experimental techniques employed in Raman spectroscopy.

A conventional Raman facility consists of five major components: (1) a source of radiation, (2) optics for illuminating the sample and collecting the Raman scattered radiation, (3) a spectrometer for dispersing the Raman scattered radiation, (4) a device for measuring the intensity of the Raman scattered light, and (5) a set of components that control the polarization of the incident radiation and monitor the polarization of the Raman scattered radiation (Chase, 1991; Ferraro and Nakamoto, 1994).

### Sources of Radiation

Prior to the introduction of lasers, the primary source of monochromatic radiation for Raman spectroscopy was the strongest excitation line in the visible region (blue, 435.83 nm) of the mercury arc. The power density of the monochromatic radiation incident on the sample from a mercury arc is very low. To compensate, relatively large samples and complicated collection optics are necessary to, respectively, generate and collect as many Raman scattered photons as possible. Collecting Raman photons that are scattered in widely different directions precludes investigating the direction and polarization characteristics of the

Raman scattered radiation. This is valuable information that is needed to identify the component(s) of the crystal's polarizability tensor that is (are) responsible for the Raman scattering (Long, 1977).

In general, laser radiation is unpolarized. However, in ionized gas lasers, such as argon and krypton ion lasers, a flat window with parallel faces and oriented at the Brewster angle ($\theta_{\text{Brewster}}$) relative to the incident beam is positioned at the end of the tube that contains the gas. This window is called a Brewster window. The Brewster angle is also called the polarizing angle and is given by the arctangent of the ratio of the indices of refraction of the two media that form the interface (e.g., for air/glass, $\theta_{\text{Brewster}} = \tan^{-1}1.5 \approx 57^\circ$). If an unpolarized beam of light traveling in air is incident on a planar glass surface at the Brewster angle, the reflected beam will be linearly polarized with its electric field vector parallel to the plane of incidence. (The plane of incidence contains the direction of propagation of the light and the normal to the interface between the two media.) A beam of light that is linearly polarized with its electric field vector in the plane of incidence and is incident on a Brewster window at $\theta_{\text{Brewster}}$ will be entirely transmitted. Thus, the light that exits from a laser tube that is capped with a Brewster window is linearly polarized (Fowles, 1989; Hecht, 1993).

Lasers have greatly increased the use of Raman spectroscopy as a tool for both research and chemical identification. In particular, continuous-wave gas lasers, such as argon, krypton, and helium-neon, provide adequately powered, monochromatic, linearly polarized, collimated, small-diameter beams that are suitable for obtaining Raman spectra from all states of matter and for a wide range of sample sizes using relatively simple systems of focusing and collection optics.

### Optics

As the light exits the laser, it generally first passes through a filter set to transmit a narrow band of radiation centered at the laser line (e.g., an interference filter). This reduces the intensity of extraneous radiation, such as the plasma lines that also exit from the laser tube. A set of mirrors and/or lenses will make the laser light incident on the sample at the desired angle and spot size.

The focusing lens reduces the beam's diameter to

$$d = \frac{4\lambda f}{\pi D} \tag{19}$$

where $\lambda$ is the wavelength of the laser radiation, $f$ is the focal length of the lens, and $D$ is the diameter of the unfocused laser beam. Note that for a given focusing lens and incident light, the size of the focused spot is inversely proportional to the size of the unfocused beam. Thus, a smaller spot size can be generated by first expanding the laser beam to the size of the focusing lens. The distance $l$ over which the beam is focused is proportional to the square of the focused beam's diameter (Hecht, 1993):

$$l = \frac{16\lambda f^2}{\pi D^2} = \frac{\pi d^2}{\lambda} \tag{20}$$

Thus, to reduce the diameter of the focal spot from an argon laser ($\lambda = 514.5$ nm) to 1 μm, the distance between the focusing lens and the sample must be accurate to within $\pm\frac{1}{2}\pi(1\ \mu\text{m})^2/(0.5145\ \mu\text{m}) \approx 3\ \mu\text{m}$.

The specularly reflected light is of no interest in Raman spectroscopy. The inelastically scattered light is collected using a fast lens (i.e., low $f$ number, defined as the focal length per diameter of the lens). Often, the lens from a 35-mm camera is a very cost-effective collection lens. The collection lens collimates the Raman scattered light and transmits it toward a second lens that focuses the light into the entrance slit of the spectrometer. The $f$ number of the second lens should match that of the spectrometer. Otherwise, the spectrometer's grating will either not be completely filled with light, which will decrease the resolving power (which is directly proportional to the size of the grating), or be smaller than the beam of radiation, which will result in stray light that may reflect off surfaces inside the spectrometer and reduce the signal-to-noise ratio.

There are many sources of stray light in a Raman spectrum. One of the major sources is elastically scattered light (it has the same wavelength as the incident radiation), caused by Rayleigh scattering by the molecules/atoms responsible for Raman scattering and Mie scattering from larger particles such as dust. Rayleigh scattering always accompanies Raman scattering and is several orders of magnitude more intense than Raman scattering. The Rayleigh scattered light comes off the sample at all angles and is captured by the collection optics along with the Raman scattered light and is directed into the spectrometer. This would not be a problem if the Rayleigh scattered light exited the spectrometer only at the 0-$\text{cm}^{-1}$ position. However, because of imperfections in the gratings and mirrors within the spectrometer, a portion of the Rayleigh scattered light exits the spectrometer in the same region as Raman scattered radiation that is shifted in the range of 0 to $\pm 250\ \text{cm}^{-1}$ from the incident radiation. Even though only a fraction of the Rayleigh scattered light is involved in this misdirection, the consequence is significant because the intensity of the Rayleigh component is so much greater than that of the Raman component. Raman peaks located within $\approx 250\ \text{cm}^{-1}$ of the laser line can only be detected if the intensity of the Rayleigh scattered light is strongly reduced.

The intensity of stray light that reaches the detector can be diminished by passing the collimated beam of elastically and inelastically scattered light collected from the sample into a notch filter. After exiting the notch filter, the light enters a lens that focuses the light into the entrance of the spectrometer.

A notch filter derives its name from the narrow band of wavelengths that it filters out. In a Raman experiment, the notch filter is centered at the exciting laser line and is placed in front of the spectrometer. (Actually, as mentioned above and explained below, the notch filter should be located in front of the lens that focuses the scattered light into the spectrometer.) The notch filter then removes the Rayleigh scattered radiation (as well as Brillouin scattered light and Mie scattering from dust particles). Depending on the width of the notch, it may also remove a significant amount of the Raman spectrum.

Dielectric notch filters consist of multiple, thin layers of two materials with different indices of refraction. The materials are arranged in alternating layers so that the variation of index of refraction with distance is a square wave. Typically, dielectric filters have a very wide notch with diffuse edges and have nonuniform transmission in the region outside the notch. As a consequence, a dielectric notch filter removes too much of the Raman spectrum (the low-wavenumber portion) and distorts a significant fraction of the portion it transmits. In contrast, holographic notch filters have relatively narrow widths, with sharp edges and fairly uniform transmission in the region outside the notch (Carrabba et al., 1990; Pelletier and Reeder, 1991; Yang et al., 1991; Schoen et al., 1993).

A holographic notch filter (HNF) is basically an interference filter. In one manufacturing process (Owen, 1992), a photosensitive material consisting of a dichromate gelatin film is placed on top of a mirror substrate. Laser light that is incident on the mirror surface interferes with its reflected beam and forms a standing-wave pattern within the photosensitive layer. The angle between the incident and reflected beams determines the fringe spacing of the hologram. If the incident beam is normal to the mirror, the fringe spacing will be equal to the wavelength of the laser light.

Chemical processing generates within the hologram-exposed material an approximately sinusoidal variation of index of refraction with distance through the thickness of the layer. The wavelength of the modulation of the index of refraction determines the central wavelength of the filter. The amplitude of the modulation and the total thickness of the filter determine the bandwidth and optical density of the filter.

The precise shape and location of the "notch" of the filter is angle tunable so the filter should be located in between the lens that collects and collimates the scattered light and the lens that focuses the Raman scattered radiation into the spectrometer such that only collimated light passes through the filter.

A holographically generated notch filter can have a relatively narrow, sharp-edged band of wavelengths that will not be transmitted. The band is centered at the wavelength of the exciting laser line and may have a total width of $\approx$300 $cm^{-1}$. The narrow width of the notch and its high optical density permit the measurement of both the Stokes and anti-Stokes components of a spectrum, as is illustrated in the Raman spectrum of zirconia presented in Figure 4.

In addition to a notch filter, it is also possible to holographically generate a bandpass filter, which can be used to remove the light from the plasma discharge in the laser tube (Owen, 1992). In contrast to dielectric filters, holographic filters can have a bandpass that is five times more narrow and can transmit up to 90% of the laser line. Dielectric bandpass filters typically transmit only 50% of the laser line.

### Optical Alignment

Alignment of the entire optical system is most easily accomplished by use of a second laser, which need only have an intensity of $\approx$1 mW. The spectrometer is set to transmit the radiation of the second laser, which is located at the exit slit of the spectrometer. For convenience, instead of having to remove the light detector, the light from the second laser can enter through a second, nearby port and made to hit a mirror that is inserted inside the spectrometer, just in front of the exit slits. The light reflected from the mirror then passes through the entire spectrometer and exits at the entrance slits, a process termed "back illumination." The back-illuminated beam travels the same path through the spectrometer (from just in front of the exit slits to the entrance slits) that the Raman scattered radiation from the sample will travel in getting to the light detector. Consequently, the system is aligned by sending the back-illuminated beam through the collection optics and into coincidence with the exciting laser beam on the sample's surface.

### Spectrometer

The dispersing element is usually a grating, and the spectrometer may typically have one to three gratings. Multiple gratings are employed to reduce the intensity of the Rayleigh line. That is, light that is dispersed by the first grating is collimated and made incident on a second grating. However, the overall intensity of the dispersed light that exits from the spectrometer decreases as the number of gratings increases. The doubly dispersed light has a higher signal-to-noise ratio, but its overall intensity is significantly lower than singly dispersed light. Since the intensity of Raman scattered radiation is generally very weak (often by one or more orders of magnitude), there is an advantage to decreasing the intensity of the Rayleigh light that enters the spectrometer by using a single grating in combination with a notch filter rather than by using multiple gratings (see Optics).

Modern spectrometers generally make use of interference or holographic gratings. These are formed by exposing photosensitive material to the interference pattern produced by reflecting a laser beam at normal incidence off a mirror surface or by intersecting two coherent beams of laser light. Immersing the photosensitized material into a suitable solvent, in which the regions exposed to the maximum light intensity experience either enhanced or retarded rates of dissolution, produces the periodic profile of the grating. The surface of the grating is then typically coated with a thin, highly reflective metal coating (Hutley, 1982).

The minimum spacing of grooves that can be generated by interference patterns is directly proportional to the wavelength of the laser light. The minimum groove spacing that can be formed using the 458-nm line of an argon ion laser is 0.28 μm (3500 grooves/mm). The line spacing of a grating dictates one of the most important characteristics of the grating, its resolving power.

The resolving power of a grating is a measure of the smallest change of wavelength that the grating can resolve:

$$\frac{\lambda}{\Delta\lambda} = \frac{mW}{d} \tag{21}$$

where $\lambda$ is the wavelength at which the grating is operating, $m$ is the diffraction order, $W$ is the width of the grating, and $d$ is the spacing of the grating grooves (or steps). This relation indicates the importance of just filling the grating with light and the influence of the spacing of the grating grooves, which is generally expressed in terms of the number of grooves per millimeters, on its resolving power.

A second parameter of interest for characterizing the performance of a grating is its absolute efficiency, which is a measure of the fraction of the light incident on the grating that is diffracted into the required order. It is a function of the shape of the groove, the angle of incidence, the wavelength of the incident light, the polarization of the light, the reflectance of the material that forms the grating, and the particular instrument that houses the gratings (Hutley, 1982). The efficiency of a grating can be measured or calculated with the aid of a computer. The measurement of the absolute efficiency is performed by taking the ratio of the flux in the diffracted beam to the flux in the incident beam. Generally, it is not necessary for a Raman spectroscopist to measure and/or calculate the absolute efficiencies of gratings. Such information may be available from the manufacturer of the gratings.

### Measurement of the Dispersed Radiation

A device for measuring the intensity of the dispersed radiation will be located just outside the exit port of the spectrometer. Since the intensity of the Raman scattered radiation is so weak, a very sensitive device is required to measure it. Three such devices are a photomultiplier tube (PMT), a photodiode array, and a charge-transfer device.

The key elements in a PMT are a photosensitive cathode, a series of dynodes, and an anode (Long, 1977; Ferraro and Nakamoto, 1994). The PMT is located at the exit of the spectrometer, which is set to transmit radiation of a particular wavelength. For Stokes-shifted Raman spectroscopy, the wavelength is longer than that of the exciting laser. As photons of this energy exit the spectrometer, they are focused into the photocathode of the PMT. For each photon that is absorbed by the photocathode, an electron is ejected and is accelerated toward the first dynode, whose potential is $\approx$100 V positive with respect to the photocathode. For each electron that hits the first dynode, several are ejected and are accelerated toward the second dynode. A single photon entering the PMT may cause a pulse of $10^6$ electrons at the anode. Long (1977) describes the different procedures for relating the current pulse at the anode to the intensity of the radiation incident on the PMT.

The efficiency of the PMT is increased through the use of a photoemission element consisting of a semiconductor whose surface is coated with a thin layer ($\approx$2 nm) of material with a low work function (e.g., a mixture of cesium and oxygen; Fraser, 1990). For a heavily doped $p$-type semiconductor coated with such a surface layer, the electron affinity in the bulk semiconductor is equal to the difference between its band-gap energy and the surface layer work function. If the work function is smaller than the band gap, the semiconductor has a negative electron affinity. This means that the lowest energy of an electron in the conduction band in the bulk of the semiconductor is higher than the energy of an electron in vacuum and results in an increase in the number of electrons emitted per absorbed photon (i.e., increased quantum efficiency). The gain of the PMT can be increased by treating the dynodesurface in a similar fashion. A dynode with negative electron affinity emits a greater number of electrons per incident electron. The overall effect is a higher number of electrons at the anode per photon absorbed at the cathode.

It is important to note that the PMT needs to be cooled to reduce the number of thermally generated electrons at the photocathode and dynodes, which add to the PMT "dark count." Heat is generally extracted from the PMT by a thermoelectric cooler. The heat extracted by the cooler is generally conducted away by flowing water.

A PMT is well suited to measuring the weak signal that exits a spectrometer that has been set to pass Raman scattered radiation of a single energy. The entire Raman spectrum is measured by systematically changing the energy of the light that passes through the spectrometer (i.e., rotating the grating) and measuring its intensity as it exits from the spectrometer. For a double monochromator fitted with a PMT, it may take minutes to tens of minutes to generate a complete Raman spectrum of one sample.

In a PMT, current is generated as photons hit the cathode. In contrast, each active segment in a photodiode array (PDA) and a charge-coupled device (CCD) stores charge (rather than generating current) that is created by photons absorbed at that location. The Raman spectrum is generated from the spatial distribution of charge that is produced in the device. In both a PDA and CCD, photons are absorbed and electron-hole pairs are created. The junction in which the electron-hole pairs are created are different in the two devices. In a PDA, the junction is a reverse-biased $p$-$n$ junction. In a CCD, the junction is the depletion zone in the semiconductor at the semiconductor-oxide interface of a metal-oxide-semiconductor.

When the reverse-biased $p$-$n$ junction is irradiated by photons with an energy greater than the band gap, electron-hole pairs are generated from the absorbed photons. Minority carriers from the pairs formed close (i.e., within diffusion distance) to the charge-depleted layer of the junction are split apart from their complementary, oppositely charged particles and driven in the opposite direction by the electric field in the junction. The increase in reverse saturation current density is proportional to the light intensity hitting the photodiode. To minimize the "dark counts", the photodiode must be cooled to reduce the number of thermally generated electron-hole pairs.

If the junction is at equilibrium and is irradiated with photons of energy greater than the band gap, electron-hole pairs generated in the junction are separated by the built-in electric field. The separated electrons and holes lower the magnitude of the built-in field. If an array of diodes is distributed across the exit of the spectrometer, the distribution of charge that is created in the array provides a measure of the intensity of the radiation that is dispersed by the spectrometer across the focal plane for the exiting radiation.

The time to measure a Raman spectrum can be greatly lowered by measuring the entire spectrum at once, rather than one energy value at a time, as is the case with a PMT. Multichannel detection is accomplished by, e.g., placing a PDA at the exit of a spectrometer. There are no slits at the exit, which is filled by the PDA. Each tiny diode in the array measures the intensity of the light that is dispersed to that location. Collectively, the entire array measures the whole spectrum (or a significant fraction of the whole spectrum) all at once. Generally, an individual photodiode in a PDA is not as good a detector as is a PMT. However, for some experiments the multichannel advantage compensates for the lower quality detector. Unfortunately, a PDA do not always provide the sensitivity at low light intensities that are needed in Raman spectroscopy.

A CCD is a charge-transfer device that combines the multichannel detection advantage of a PDA with a sensitivity at low light intensities that rivals that of a PMT (Bilhorn et al., 1987a,b; Epperson, 1988). Charge-coupled devices are made of metal-oxide-semiconductor elements and, in terms of light detection, function analogously to photographic film. That is, photons that hit and are absorbed at a particular location of the CCD are stored at that site in the form of photogenerated electrons (for *p*-type semiconductor substrate) or photogenerated holes (for *n*-type semiconductor substrate).

One example of a CCD is a heavily doped, *p*-type silicon substrate whose surface is coated with a thin layer of $SiO_2$ on top of which are a series of discrete, closely spaced metal strips, referred to as gates. The potential of each adjacent metal strip is set at a different value so that the width of the depletion layer in the semiconductor varies periodically across its surface. When a positive potential is applied to a metal strip, the mobile holes in the *p*-type silicon are repelled from the surface region, creating a depleted layer. If a high enough potential is applied to a metal strip, significant bending of the bands in the region of the semiconductor close to the oxide layer causes the bottom of the conduction band to approach the Fermi level. Electrons occupy states in the conduction band near the surface forming an inversion layer, i.e., an *n*-type surface region in the bulk *p*-type semiconductor. Photons that are absorbed in the surface region generate electron-hole pairs in which the electrons and holes are driven apart by the field in the depletion layer. The holes are expelled from the depletion layer and the electrons are stored in the potential well at the surface. As the intensity of light increases, the number of stored electrons increases. The light exiting the spectrometer is dispersed and so hits the CCD at locations indicative of its wavelength. The dispersed light creates a distribution of charge in the depletion layer from which the sample Raman spectrum is generated. The distribution of charge corresponding to the Raman spectrum is sent to a detector by shifting the periodic variation in potential of the metal strips in the direction of the detector. In this manner the charge stored in the depletion layer of the semiconductor adjacent to a particular metal strip is shifted from one strip to the next. Charge arrives at the detector in successive packets, corresponding to the sequential distribution of the metal gates across the face of the CCD.

An illustration of the increased efficiency of measuring Raman spectra that has emerged in the past 10 years is provided by a comparison of the time required to measure a portion (200 to 1500 $cm^{-1}$) of the surface-enhanced Raman spectrum (see the following paragraph) of thin passive films grown on samples of iron immersed in aqueous solutions. Ten to 15 min (Gui and Devine, 1991) was required to generate the spectrum by using a double spectrometer (Jobin-Yvon U1000 with 1800 grooves/mm holographic gratings) and a PMT (RCA c31034 GaAs), while the same spectrum can now be acquired in $\approx$5 s (Oblonsky and Devine, 1995) using a single monochromator (270M Spex), in which the stray light is reduced by a notch filter (Kaiser Optics Super Holographic Notch Filter centered at 647.1 nm), and the intensity of the dispersed Raman scattered radiation is measured with a CCD (Spectrum One, 298 $\times$ 1152 pixels). The enhanced speed in generating the spectrum makes it possible to study the time evolution of samples.

In the previous paragraph, mention was made of surface-enhanced Raman scattering (SERS), which is the greatly magnified intensity of the Raman scattered radiation from species adsorbed on the surfaces of specific metals with either roughened surfaces or colloidal dimensions (Chang and Furtak, 1982). SERS is not a topic of this unit, but a great deal of interest during the past 15 years has been devoted to the use of SERS in studies of surface adsorption and surface films.

### Polarization

By controlling the polarization of the incident radiation and measuring the intensity of the Raman scattered radiation as a function of its polarization, it is possible to identify the component(s) of the polarizabilty tensor of a crystal that are responsible for the different peaks in the Raman spectrum (see Example: Raman Active Vibrational Modes of $\alpha$-$Al_2O_3$, below). This information will then identify the vibrational mode(s) responsible for each peak in the spectrum. The procedure is illustrated below for $\alpha$-$Al_2O_3$ (see Example: Raman Active Vibrational Modes of $\alpha$-$Al_2O_3$). The configuration of the polarizing and polarization measuring components are discussed in Scherer (1991).

### Fourier Transform Raman Spectroscopy

The weak intensity of Raman scattering has already been mentioned as one of the major shortcomings of Raman spectroscopy. Sample heating and fluorescence are two other phenomena that can increase the difficulty of obtaining Raman spectra. Sample heating is caused by absorption of energy during illumination with laser beams of high power density. Fluorescence is also caused by absorption of laser energy by the sample (including impurities in the sample). When the excited electrons drop back down to lower energy levels, they emit light whose energy equals the difference between the energies of the initial (excited) and final states. The fluorescence spectrum is (practically speaking) continuous and covers a wide range of values. The intensity of the fluorescence can be much greater than that of the Raman scattered radiation.

Both sample heating and fluorescence can be minimized by switching to exciting radiation whose energy is too low to be absorbed. For many materials, optical radiation in the red or near-IR range will not be absorbed. If such radiation is to be used in Raman spectroscopy, two consequences must be recognized and addressed. First, since Raman scattering is a light-scattering process, its intensity varies as $\lambda^{-4}$. Hence, by switching to longer wavelength exciting radiation, the intensity of the Raman spectrum will be significantly lowered. Second, the sensitivity of most PMTs to red and especially near-infrared radiation is very low. Consequently, the use of long-wavelength radiation in Raman spectroscopy has been coupled to the use of Fourier transform Raman spectroscopy (FTRS) (Parker, 1994).

In FTRS, an interferometer is used in place of a monochromator. The Raman scattered radiation is fed directly into a Michelson interferometer and from there it enters the detector. The Raman spectrum is obtained by taking the cosine-Fourier transform of the intensity of the radiation that reaches the detector, which varies in magnitude as the path difference between the moving mirror and the fixed mirror within the interferometer changes. Since the Raman scattered radiation is not dispersed, a much higher Raman intensity reaches the detector in FTRS than is the case for conventional Raman spectroscopy. The much higher intensity of the Raman scattered radiation in FTRS permits the use of longer wavelength incident radiation (e.g., $\lambda = 1064$ nm of Nd-YAG), which may eliminate fluorescence.

## DATA ANALYSIS AND INITIAL INTERPRETATION

Raman scattering may result from incident radiation inducing transitions in electronic, vibrational, and rotational states of the scattering medium. Peaks in the Raman spectrum of a solid obtained using visible exciting radiation are generally associated with vibrational modes of the solid. The vibrations may be subdivided into internal modes that arise from molecules or ions that make up the solid and external modes that result from collective modes ($k = 0$) of the crystal. In either case, the presence of a peak in the Raman spectrum due to the vibration requires that the Raman transition be symmetry allowed, i.e., that the integrand in Equation 7 have a nonzero value. In addition, for a measurable Raman intensity, the integral in Equation 7 must have a large enough magnitude. It is generally difficult to calculate the intensity of a Raman peak, but it is rather straightforward to use group theoretical techniques to determine whether or not the vibrational mode is symmetry allowed.

### Example: Raman Active Vibrational Modes of $\alpha$-$Al_2O_3$

The extremely useful result that the crystallographic point group provides all the information that is needed to know the symmetries of the Raman active vibrational modes of the crystal is derived below (see Appendix). The left-hand column in the character table for each point group lists a set of symbols such as $A_1$, $B_1$, and $B_2$ for the point group $C_{2v}$. The term $A_1$ is the list of characters of the matrices that represent the effect of each symmetry operation of the point group on any function listed in the far right-hand column of the character table. A remarkable result of group theory is that the list of characters of the matrices that represent the effect of each symmetry operation of the point group on "any" function can be completely described by a linear combination of the few lists presented in the character table. For example, for the point group $C_{2v}$, the symmetry of the function $x^2$ is described by $A_1$. This means that when the effect of each of the symmetry operations operating on $x^2$ is represented by a matrix, the characters of the four matrices are 1, 1, 1, 1. In fact, within the context of the point group $C_{2v}$, the symmetry of any function is given by $aA_1 + bB_1 + cB_2$, where $a$, $b$, and $c$ are integers. In the language of group theory, $A_1$, $B_1$, and $B_2$ are called the irreducible representations of the point group $C_{2v}$. The linear combination of irreducible representations that describe the symmetry of a function is called the symmetry species of the function.

The irreducible representations of the normal vibrational modes of a crystal are most easily identified by the correlation method (Fately et al., 1971). The analysis for the lattice vibrations of $\alpha$-$Al_2O_3$ will be summarized here. The details of the analysis are contained in Fately et al. (1971).

The space group of $\alpha$-$Al_2O_3$ is $D^6_{3d}$. There are two molecules of $\alpha$-$Al_2O_3$ per Bravais cell. Each atom in the Bravais cell has its own symmetry, termed the site symmetry, which is a subgroup of the full symmetry of the Bravais unit cell. The site symmetry of the aluminum atoms is $C_3$ and that of the oxygen atoms is $C_2$. From the character table for the $C_3$ point group, displacement in the $z$ direction is a basis for the $A$ irreducible representation. Displacements parallel to the $x$ and $y$ axes are bases for the $E$ irreducible representation. The correlation tables for the species of a group and its subgroups (Wilson et al., 1955) correlate the $A$ species of $C_3$ to $A_{1g}$, $A_{2g}$, $A_{1u}$, and $A_{2u}$ of the crystal point group $D_{3d}$.

Displacement of the oxygen atom in the $z$ direction is a basis of the $A$ irreducible representation of the $C_2$ point group, which is the site symmetry of the oxygen atoms in $\alpha$-$Al_2O_3$. Displacements of the oxygen atom that are parallel to the $x$ and $y$ axes are bases for the $B$ irreducible representation of $C_2$. The correlation table associates the $A$ species of $C_2$ to $A_{1g}$, $E_g$, $A_{1u}$, and $E_u$ of $D_{3d}$. The $B$ species of $C_2$ correlates to $A_{2g}$, $E_g$, $A_{2u}$, and $E_u$ of $D_{3d}$.

After accounting for the number of degrees of freedom contributed by each irreducible representation of $C_2$ and $C_3$ to an irreducible representation of $D_{3d}$ and removing the irreducible representations associated with the rigid translation of the entire crystal, the irreducible species for the optical modes of the corundum crystal are determined to be

$$2A_{1g} + 2A_{1u} + 3A_{2g} + 2A_{2u} + 5E_g + 4E_u \qquad (22)$$

Consulting the character table of the point group $D_{3d}$ indicates that the Raman active vibrational modes would span

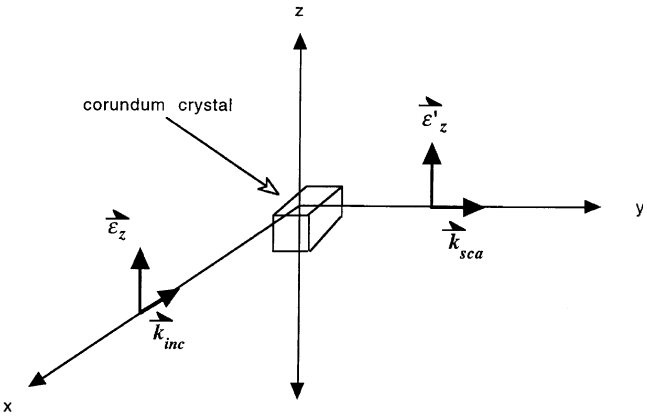



**Figure 8.** Directions and polarizations of incident and scattered radiation for the $x(zz)y$ configuration.

the symmetry species $A_{1g}$ and $E_g$. The basis functions for these symmetry species are listed in the character table:

$$x^2 + y^2 \quad \text{and} \quad z^2 \quad \text{for } A_{1g} \tag{23}$$

and

$$(x^2 - y^2, xy) \quad \text{and} \quad (xz, yz) \quad \text{for } E_g \tag{24}$$

Thus there are a total of seven Raman active modes so there may be as many as seven peaks in the Raman spectrum. The question now is which peaks correspond to which vibrational modes.

Figure 8 depicts a Cartesian coordinate system, and a crystal of corundum is imagined to be located at the origin with the crystal axes parallel to $x$, $y$, and $z$. If light is incident on the crystal in the $-x$ direction and polarized in the $z$ direction, and scattered light is collected in the $y$ direction after having passed through a $y$ polarizer, then the experimental conditions of incidence and collection are described by the Porto notation (Porto and Krishnan, 1967) as $x(zz)y$. The first term inside the parentheses indicates the polarization of the incident light and the second term denotes the polarization of the scattered light that is observed. In general, if $z$ polarized light is incident on a crystal in the $x$ direction, then the induced polarization of the molecule will be given by $[p_x, p_y, p_z] = [\alpha_{xz}E_z, \alpha_{yz}E_z, \alpha_{zz}E_z]$. Light scattered in the $y$ direction would be the result of the induced polarization components $p_x$ and $p_z$. If the light scattered in the $y$ direction is passed through a $z$ polarizer, then the source of the observed radiation would be the induced polarization along the $z$ direction. Consequently, only the $\alpha_{zz}$ component of the polarizability contributes to scattered light in the case of $x(zz)y$. Thus, peaks present in the Raman spectrum must originate from Raman scattering by vibrational modes that span the $A_{1g}$ irreducible species.

Figure 9 presents the Raman spectra reported by Porto and Krishnan (1967) for single-crystal corundum under a variety of incidence and collection conditions. Figure 9A is the Raman spectrum in the $x(zz)y$ condition that was discussed above so the two peaks present in the spectrum correspond to vibrational modes with $A_{1g}$ symmetry. The peaks present in the spectrum in Figure 9B can originate from vibrational modes with either $A_{1g}$ or $E_g$ symmetries. Consequently, the peaks present in Figure 9B but missing in Figure 9A have $E_g$ symmetry. The spectra in Figures 9C-E exhibit only peaks with $E_g$ symmetry. Collectively, the spectra reveal all seven Raman active modes and distinguish between the $A_{1g}$ modes and the $E_g$ modes.

The magnitude of the Raman shift in wavenumbers of each peak with respect to the exciting laser line can be related to the energy of the vibration. Combined with the knowledge of the masses of the atoms involved in the vibrational mode and, e.g., the harmonic oscillator assumption, it is possible to calculate the force constant associated with the bond between the atoms participating in the vibration. If one atom in the molecule were replaced by another without changing the symmetry of the molecule, the peak would shift corresponding to the different strength of the bond and the mass of new atom compared to the original atom. This illustrates why the Raman spectrum can provide information about bond strength and alloying effects.

### Other Examples of the Use of Raman Spectroscopy in Studies of Solids

Raman spectroscopy can provide information on the amorphous/crystalline nature of a solid. Many factors can contribute to the widths of peaks in the Raman spectra of a solids. The greatest peak broadening factor can be microcrystallinity. For a macroscopic-sized single crystal of silicon at room temperature, the full width at half-maximum (FWHM) of the peak at 522 $cm^{-1}$ is $\approx$3 $cm^{-1}$ (Pollak, 1991). The breadth of the peak increases dramatically as the size of the crystal decreases below 10 nm. The cause of peak broadening in microcrystals is the finite size of the central peak in the Fourier transform of a phonon in a finite-size crystal. In an infinitely large crystal, the Fourier transform of a phonon consists of a single sharp line at the phonon frequency. The peak width $\Delta k$ (i.e., FWHM) in a crystal of dimension $L$ is given approximately by

$$\Delta k = \frac{2\pi(v/c)}{L} \tag{25}$$

where $v$ is phonon velocity and $c$ the velocity of light. Assuming $v = 1 \times 10^5$ cm/s, $\Delta k = 20$ $cm^{-1}$ for $L = 10$ nm and $\Delta k = 200$ $cm^{-1}$ for $L = 1$ nm. In the extreme case, the silicon is amorphous, in which case the $k = 0$ selection rule breaks down and all phonons are Raman active. In that case, the Raman spectrum resembles the phonon density of states and is characterized by two broad peaks centered at 140 and 480 $cm^{-1}$ (Pollak, 1991). Consequently, the Raman spectrum can distinguish between the crystalline and amorphous structures of a solid and can indicate the grain size of microcrystalline solids.

Raman spectroscopy can be used to nondestructively measure the elastic stress in crystalline samples. Elastic straining of the lattice will alter the spring constant of a chemical bond and, hence, will shift the frequency of the

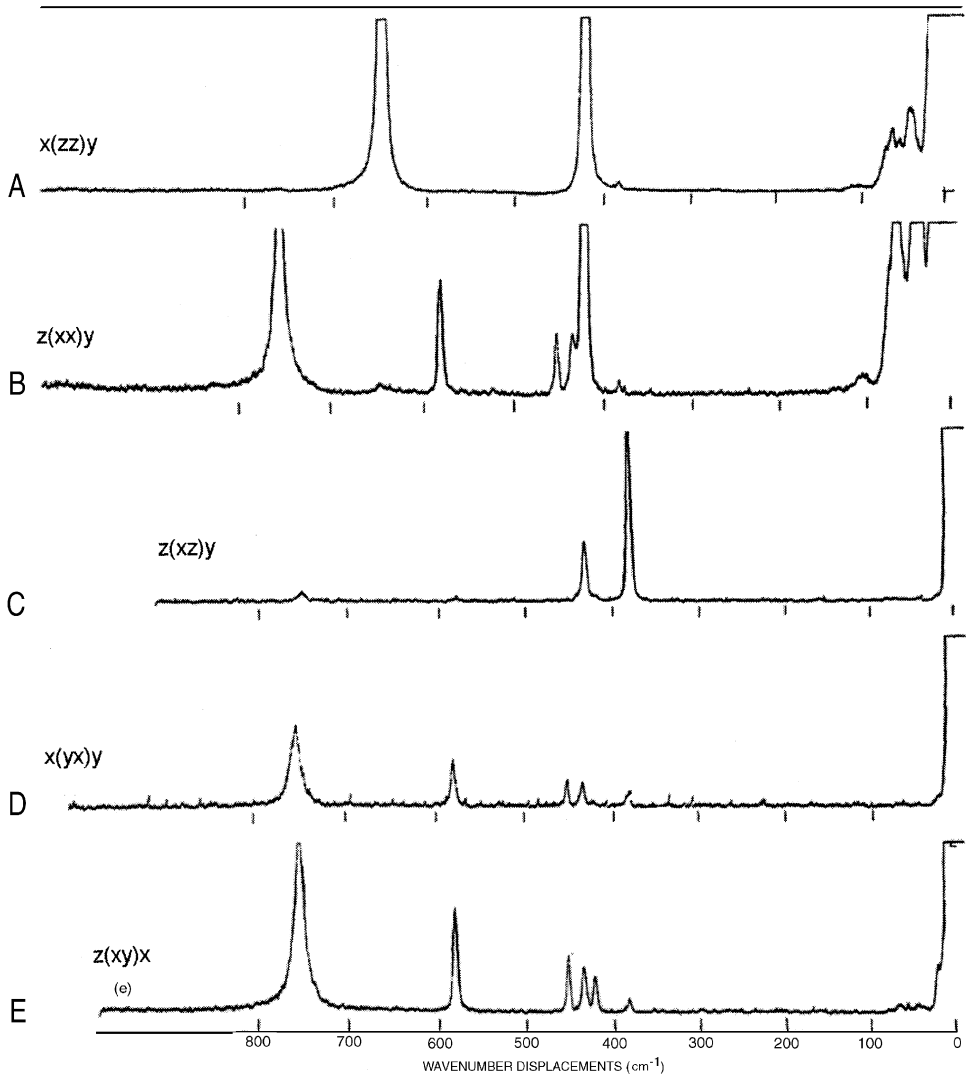



**Figure 9.** Raman spectra of corundum as a function of crystal and laser light polarization orientations (Porto and Krishnan, 1967).

vibrational mode associated with the strained bond. The direction and magnitude of the shift in peak location will depend on the sign (tensile or compressive) and magnitude, respectively, of the strain. Raman spectroscopy has been used to measure the residual stresses in thin films deposited on substrates that result in lattice mismatches and in thermally cycled composites where the strains result from differences in thermal expansion coefficients of the two materials (Pollak, 1991).

Another example of the use of Raman spectroscopy to nondestructively investigate the mechanical behavior of a solid is its use in studies of $Al_2O_3$-$ZrO_2$ composites (Clarke and Adar, 1982). These two-phase structures exhibit improved mechanical toughness due to transformation of the tetragonal $ZrO_2$ to the monoclinic form. The transformation occurs in the highly stressed region ahead of a growing crack and adds to the energy that must be expended in the propagation of the crack. Using Raman microprobe spectroscopy, it was possible to measure, with a spatial resolution of $\approx$1 μm, the widths of the regions on either side of a propagating crack in which the $ZrO_2$ transformed. The size of the transformed region is an important parameter in theories that predict the increased toughness expected from the transformation of the $ZrO_2$.

Phase transformations resulting from temperature or pressure changes in an initially homogeneous material have also been studied by Raman spectroscopy (Ferraro and Nakamoto, 1994). Raman spectra can be obtained from samples at temperatures and pressures that are markedly different from ambient conditions. All that is needed is the ability to irradiate the sample with a laser beam and to collect the scattered light. This can be accomplished by having an optically transparent window (such as silica, sapphire, or diamond) in the chamber that houses the sample and maintains the nonambient conditions. Alternatively, a fiber optic can be used to transmit the

incident light to the sample and the scattered light to the spectrometer.

The influence of composition on the Raman spectrum is in evidence in semiconductor alloys such as $Ga_{1-x}AL_xAs$ $(0 < x < 1)$, which exhibit both GaAs-like and AlAs-like longitudinal optical (LO) and transverse optical (TO) phonon modes. The LO modes have a greater dependence on composition than do the TO modes. The AlAs-like LO mode shifts by 50 $cm^{-1}$ and the GaAs-like mode shifts by $\approx$35 $cm^{-1}$ as $x$ varies from 0 to 1. Thus the Raman spectrum of $Ga_{1-x}Al_xAs$ $(0 < x < 1)$ can be used to identify the composition of the alloy (Pollak, 1991).

Similarly, the Raman spectra of mixtures of $HfO_2$ and $ZrO_2$ are strong functions of the relative amounts of the two components. Depending on the particular mode, the Raman peaks shift by $\approx$7 to $\approx$60 $cm^{-1}$ as the amount of $HfO_2$ increases from 0 to 100% spectroscopy (Ferraro and Nakamoto, 1994).

Raman spectroscopy has long been used to characterize the structure of polymers: identifying functional groups, end groups, crystallinity, and chain orientation. One word of caution: As might be expected of any Raman study of organic matter, fluorescence problems can arise in Raman investigations of polymers (Bower and Maddams, 1989; Bulkin, 1991; Rabolt, 1991).

### Raman Spectroscopy of Carbon

Because carbon can exist in a variety of solid forms, ranging from diamond to graphite to amorphous carbon, Raman spectroscopy is the single most effective characterization tool for analyzing carbon. Given the wide spectrum of structures exhibited by carbon, it provides a compelling example of the capability of Raman spectroscopy to analyze the structures of a large variety of solids. The following discussion illustrates the use of Raman spectroscopy to distinguish between the various structural forms of carbon and describes the general procedure for quantitatively analyzing the structure of a solid by Raman spectroscopy.

Figure 10, which was originally presented in an article by Robertson (1991), who compiled the data of a number of researchers, presents the Raman spectra of diamond, large crystal graphite, microcrystalline graphite, glassy carbon, and several forms of amorphous carbon. Diamond has two atoms per unit cell and therefore has a single internal vibrational mode $(3N - 5 = 1)$. This mode is Raman active and is located at 1332 $cm^{-1}$. Graphite, with four atoms per unit cell, has six internal vibrational modes $(3N - 6 = 6)$, two of which are Raman active. The rigid-layer mode of graphite spans the symmetry species $E_{2g}$ and occurs at 50 $cm^{-1}$, which is too close to the incident laser line to be accessible with most Raman collection optics and spectrometers. The second Raman active mode of graphite is centered at 1580 $cm^{-1}$, and it too spans the symmetry species $E_{2g}$. The relative displacements of the carbon atoms during this in-plane mode are presented next to the Raman spectrum in Figure 10. Thus, Raman spectroscopy can easily distinguish between diamond and large-crystal graphite.

The third spectrum from the top of Figure 10 was obtained from microcrystalline graphite. The next lower spectrum is similar and belongs to glassy carbon. There

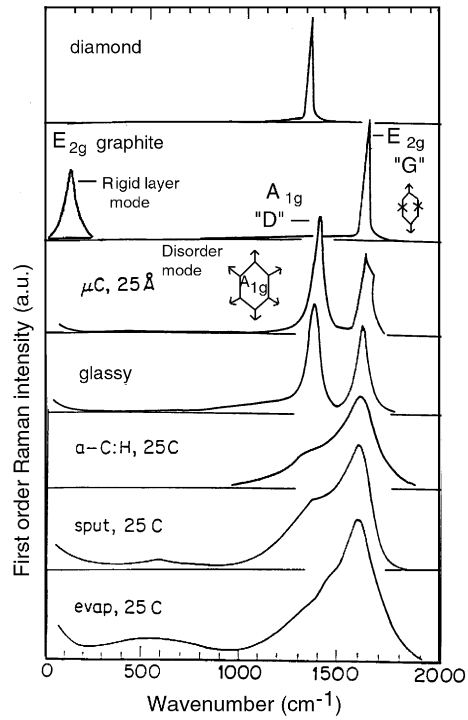



**Figure 10.** First-order Raman spectra of diamond, highly oriented pyrolytic graphite (hopg), microcrystalline graphite, glassy C, plasma-deposited a-C:H, sputtered a-C, and evaporated a-C. (From Robertson, 1991.)

are two peaks in these spectra, one at 1580 $cm^{-1}$ and the other at 1350 $cm^{-1}$. Given these peak positions, originally there was some thought that the structures were a mixture of graphitelike ($sp^2$ bonding) and diamondlike ($sp^3$ bonding) components. However, experiments that revealed the effects of heat treatments on the relative intensities of the two peaks clearly showed that the material was polycrystalline graphite and the peak at 1350 $cm^{-1}$ is a consequence of the small size of the graphite crystals. The new peak at 1350 $cm^{-1}$ is labeled the disorder, or "$D$," peak, as it is a consequence of the breakdown of the crystal momentum conservation rule. In the Raman spectrum of an infinite crystal, the peak at 1350 $cm^{-1}$ is absent. This peak results from the vibrational mode sketched next to the spectrum for microcrystalline graphite in Figure 10, and it spans the symmetry species $A_{1g}$. Its Raman activity is symmetry forbidden in a crystal of infinite size. Finite crystal size disrupts translational symmetry, and so the $D$ mode is Raman active in microcrystalline graphite. The ratio of the integrated intensities of the peaks associated with the $D$ and $G$ modes is proportional to the size of the crystal:

$$\frac{I(D\,\text{mode})}{I(G\,\text{mode})} = \frac{k}{d} \tag{26}$$

where $d$ is the crystal size. This relationship may be used, for example, to calculate the relative sizes of two microcrystals and the relative mean grain diameters of two

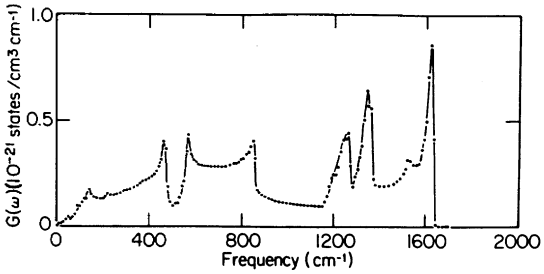



**Figure 11.** Phonon density of states for graphite.

polycrystalline aggregates, so long as the grain size is greater than 12 Å. The behavior of crystals smaller than 12 Å begins to resemble that of amorphous solids.

An amorphous solid of $N$ atoms may be thought of as a giant molecule of $N$ atoms. All $3N - 6$ of its internal vibrational modes are Raman active, and the Raman spectrum is proportional to the phonon density of states. This is demonstrated by a comparison of the bottom three spectra in Figure 10 with the calculated phonon density of states of graphite in Figure 11.

At this point several important distinctions between x-ray diffraction and Raman spectroscopy can be appreciated. Like Raman spectroscopy, x-ray diffraction is also able to distinguish between crystals, microcrystals, and amorphous forms of the same material. Raman spectroscopy not only is able to distinguish between the various forms but also readily provides fundamental information about the amorphous state as well as the crystalline state. This is due to the fact that x-ray diffraction provides information about the long-range periodicity of the arrangement of atoms in the solid, while Raman spectroscopy provides information concerning the bonds between atoms and the symmetry of the atomic arrangement. On the other hand, while x-ray diffraction is in principle applicable to every crystal, Raman spectroscopy is only applicable to those solids with unit cells containing two or more atoms.

### Quantitative Analysis

The quantitative analysis of a spectrum requires fitting each peak to an expression of Raman scattered intensity vs. frequency of the scattered light. Typically, a computer is used to fit the data to some mathematical expression of the relationship between the intensity of the Raman scattered radiation as a function of its frequency with respect to that of the incident laser line. One example of a peak-fitting relationship is the expression (DiDomenico et al., 1968)

$$\frac{dI}{d\omega} = \frac{(\text{const})\Gamma\omega_0}{\{\omega_0^2 - (\Delta\omega)^2\}^2 + 4\Gamma^2\omega_0^2(\Delta\omega)^2} \qquad (27)$$

where $dI/d\omega$ is the average Raman scattered intensity per unit frequency and is proportional to the strength of the signal at the Raman shift of $\Delta\omega$; $\Gamma$ is the damping constant, $\omega_0$ is the frequency of the undamped vibrational mode, and $\Delta\omega$ is the frequency shift from the laser line. The calculated fit yields the values of $\Gamma$ and $\omega_0$ for each peak in the spectrum.

An example of the quantitative analysis of a Raman spectrum is provided in the work of Dillon et al. (1984), who used Raman spectroscopy to investigate the growth of graphite microcrystals in carbon films as a function of annealing temperature. The data for the $D$ and $G$ peaks of graphite were fitted to Equation 27, which was then integrated to give the integrated intensity of each peak. The ratio of the integrated intensity of the $D$ mode to the integrated intensity of the $G$ mode was then calculated and plotted as a function of annealing temperature. The ratio increased with annealing temperature over the range from 400 to 600°C, and the widths of the two peaks decreased over the same range. The results suggest that either the number or the size of the crystal grains was increasing over this temperature range. The integrated intensity ratio of the $D$-to-$G$ mode reached a maximum and then decreased with higher temperatures, indicating growth in the grain size during annealing at higher temperatures. Dillon et al. also used the peak positions defined by Equation 27 to monitor the shift in peak locations with annealing temperature. The results indicated that the annealed films were characterized by threefold rather than fourfold coordination.

In summary, a single-phase, unknown material can be identified by the locations of peaks and their relative intensities in the Raman spectrum. The location of each peak and its breadth can be defined by fitting the peaks to a quantitative expression of peak intensity vs. Raman shift. The same expression will provide the integrated intensity of each peak. Ratios of integrated intensities of peaks belonging to different phases in a multiphase sample will be proportional to the relative amounts of the two phases.

## PROBLEMS

Many problems can arise in the generation of a Raman spectrum. Peaks not part of the spectrum of the sample can sometimes appear. Among the factors that can produce spurious peaks are plasma lines from the laser and cosmic rays hitting the light detector. The occurrence of peaks caused by cosmic rays increases with the time required for acquisition of the spectrum. The peaks from cosmic rays can be identified by their sharpness and their nonreproducibility. Peaks attributed to plasma lines are also sharp, but not nearly as sharp as those caused by cosmic rays. A peak associated with a plasma line will vanish when the spectrum is generated using a different laser.

An all too common problem encountered in Raman spectroscopy is the occurrence of fluorescence, which completely swamps the much weaker Raman signal. The fluorescence can be caused by the component of interest or by impurities in the sample. Fluorescence can be minimized by decreasing the concentration of the guilty impurity or by long-time exposure of the impurity to the exciting laser line, which "burns out" the fluorescence. Absorption of the exciting laser line by the impurity results in its thermal decomposition.

Fluorescence is generally red shifted from the exciting laser line. Consequently, the anti-Stokes Raman spectrum may be less affected by fluorescence than the Stokes-Raman spectrum. The intensity of the anti-Stokes Raman spectrum is much weaker than that of the Stokes Raman so this is not always a viable approach to avoiding the fluorescence signal.

Increasing the wavelength of the exciting laser line may also reduce fluorescence, as was mentioned above (see Fourier Transform Raman Spectroscopy). One potential problem with the latter approach is tied to the wavelength dependence of the intensity of the scattered light, which varies inversely as the fourth power of the wavelength. Shifting to exciting radiation with a longer wavelength may reduce fluorescence but it will also decrease the overall intensity of the scattered radiation.

Fluctuations in the intensity of the incident laser power during the generation of a Raman spectrum are particularly problematic for spectra measured using a single-channel detector. Errors in the relative intensities of different peaks can result from fluctuations in the laser power. Similarly, fluctuations in laser power will cause variations in the intensities of successive measurements of the same spectrum using a multiple-channel detector. These problems will be largely averted if the laser can be operated in a constant-light-intensity mode rather than in a controlled (electrical) power input mode.

If the sample absorbs the incident radiation, problems may occur even if the absorption does not result in fluorescence. For example, if an argon laser ($\lambda = 514.5$ nm) were used to generate the Raman spectrum of an adsorbate on the surface of a copper or gold substrate, significant reduction in the Raman intensity would result because of the strong absorption by copper and gold of green light. In this case, better results would be obtained by switching to a krypton laser ($\lambda = 647.1$ nm) or helium-neon laser ($\lambda = 632.8$ nm).

Absorption of the incident radiation can also lead to increases in the temperature of the sample. This may have a major deleterious effect on the experiment if the higher temperature causes a change in the concentration of an adsorbate or in the rate of a chemical or electrochemical reaction. If optically induced thermal effects are suspected, the spectra should be generated using several different incident wavelengths.

Temperature changes can also cause significant changes in the Raman spectrum. There are a number of sources of the temperature dependency of Raman scattering. First, the intensity of the Stokes component relative to the anti-Stokes component decreases as temperature increases. For studies that make use of only the Stokes component, an increase in temperature will cause a decrease in intensity. A second cause of the temperature dependency of a Raman spectrum is the broadening of peaks as the temperature increases. This cause of peak broadening is dependent on the decay mechanism of the phonons. Broadening of the Raman peak can also result from hot bands. Here the higher temperature results in a higher population of the excited state. As a result, the incident photon can induce the transition from the first excited state to the second excited state. Because of anharmonic effects, the energy required for this transition is different from the energy required of the transition from the ground state to the first excited state. As a result, the peak is broadened. Hence, Raman peaks are much sharper at lower temperatures. Consequently, it may be necessary to obtain spectra at temperatures well below room temperature in order to minimize peak overlap and to identify distinct peaks.

Temperature changes can also affect the Raman spectrum due to the temperature dependencies of the phonons themselves. For example, the phonon itself will have a strong temperature dependency as the temperature is raised close to that of a structural phase transition if at least one component of the oscillation coincides with the displacement during the phase transition. Temperature changes can also affect the phonon frequency. As the temperature increases, the anharmonic character of the vibration causes the well in the plot of potential energy vs. atomic displacement to be more narrow than is the case for a purely harmonic oscillation. The atomic displacements are therefore smaller, which results in a higher frequency since the amplitude of displacement is inversely proportional to frequency. In addition, thermal expansion increases the average distance between the atoms as temperature increases, leading to a decrease in the strength of the interatomic interactions. This results in a decrease in the phonon frequency with increasing temperature. In summary, superior Raman spectra are generally obtained at lower temperatures, and it may be necessary in some cases to obtain spectra at temperatures that are considerably lower than room temperature.

Changes in the temperature of the room in which the Raman spectra are measured can cause changes in the positions of optical components (lenses, filters, mirrors, gratings) both inside and outside the spectrometer. Such effects are important when attempting to accurately measure the locations of peaks in a spectrum, or, e.g., stress-induced shifts in peak locations.

One of the more expensive mistakes that can be made in systems using single-channel detectors is the inadvertent exposure of the PMT to the Rayleigh line. The high intensity of the Rayleigh scattered radiation can "burn out" the PMT, resulting in a significant increase in PMT dark counts. Either closing the shutter in front of the PMT or shutting off the high voltage to the PMT will prevent damage to the PMT from exposure to high light intensity. Typically, accidents of this type occur when the frequency of the incident radiation is changed and the operator neglects to note this change at the appropriate point in the software that runs the experiment and controls the operation of the PMT shutter.

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*In Chapter 11, Bloch sums are used for the eigenvectors of a crystal's vibrational Hamiltonian.*

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*Provides a highly readable introduction to quantum mechanics and group theory.*

Fateley et al., 1971. See above.

*Recipe for determining the symmetry species of vibrational modes of crystals.*

Ferraro and Nakamoto, 1994. See above.

*Provides a more current description of equipment and more extensive and up-to-date discussion of applications of Raman spectroscopy than are provided in the text by Long (see below).*

Long, 1977. See above.

*Provides a thorough introduction to Raman spectroscopy.*

Pollak, 1991. See above.

*A comprehensive review of the use of Raman spectroscopy to investigate the structure and composition of semiconductors.*

## APPENDIX: GROUP THEORY AND VIBRATIONAL SPECTROSCOPY

This appendix was written, in part, to provide a road map that someone interested in vibrational spectroscopy might want to follow in migrating through the huge number of topics that are covered by the vast number of textbooks on group theory (see Literature Cited). Although necessarily brief in a number of areas, it does cover the following topics in some depth, selected on the basis of their importance and/or the rareness with which they are fully discussed elsewhere: (1) How character tables are used to identify the Raman activity of vibrational modes is demonstrated. (2) That the $ij$ component, $\alpha_{ij}$, of the polarizability tensor has the same symmetry as the product function $x_i x_j$ is demonstrated. Character tables list quadratic functions such as $x^2$, $yz$ and identify the symmetry species (defined below) spanned by each. The same symmetry species will be spanned by the Raman active vibrational modes. (3) The proper rotations found in crystallographic point groups are identified. (4) A general approach, developed by Lax (1974) and using multiplier groups, is presented that elegantly establishes the link between all space groups, both symmorphic and nonsymmorphic, and the crystallographic point groups.

### Vibrational Selection Rules

Quantum mechanically a vibrational Raman transition may occur if the integral in Equation 7 has a nonzero value. This can quickly be determined by the use of group theoretical techniques, which are summarized in the form of a Raman selection rule. The Raman selection rule is a statement of the symmetry that a vibrational mode must possess in order that it may be Raman active.

Before discussing the symmetry-based selection rules for Raman scattering, there is also a restriction based on the principles of energy and momentum conservation

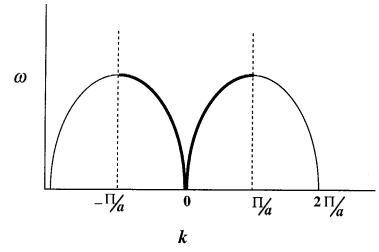



**Figure 12.** Dispersion of longitudinal wave in a linear monoatomic lattice (nearest-neighbor interactions only).

that is especially helpful in considering the Raman activity of solids. The requirements of energy and momentum conservation are expressed as follows:

$$\hbar\omega_i = \hbar\omega_s \pm \hbar\omega_v \tag{28}$$

$$\hbar k_i = \hbar k_s \pm \hbar k_v \tag{29}$$

where $\hbar k_i, \hbar k_s$, and $\hbar k_v$ are the momentum vectors of the incident and scattered radiation and the crystal phonon, respectively. For optical radiation $|k_i|$ is on the order of $10^4$ to $10^5$ cm$^{-1}$. For example, for $\lambda = 647.1$ nm, which corresponds to the high-intensity red line of a krypton ion laser,

$$k = \frac{2\pi}{\lambda} = 9.71 \times 10^3 \text{ cm}^{-1} \tag{30}$$

For a crystal with $a_0 = 0.25$ nm, $k_{v,\max} \approx 2.51 \times 10^8$ cm$^{-1}$. Hence, for Raman scattering, $k_v \ll k_{v,\max}$. In other words, $k_v$ must be near the center of the Brillouin Zone (BZ).

The phonon dispersion curves for a simple, one-dimensional, monoatomic lattice and a 1D diatomic lattice are presented in Figures 12 and 13. Since there are no available states at the center of the BZ in a monatomic lattice [for a three-dimensional (3D) lattice as well as a one-dimensional lattice], such structures are not Raman active. Consequently, all metals are Raman inactive. Diatomic lattices, whether homonuclear or heteronuclear, do possess phonon modes at the center of the BZ (see Fig. 13) and are Raman active. Thus, crystals such as diamond, silicon, gallium nitride, and aluminum oxide are all Raman active.

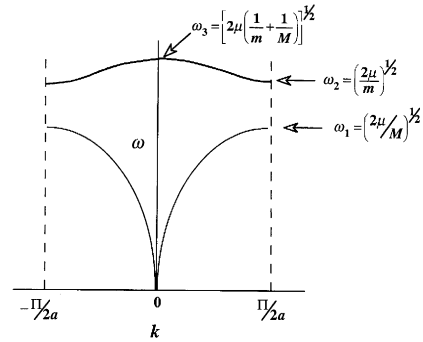



**Figure 13.** Dispersion of longitudinal wave in a linear diatomic lattice.

A Raman spectrum may consist of several peaks in the plot of the intensity of the scattered radiation versus its shift in wavenumber with respect to the incident radiation, as illustrated in Figure 4. If the immediate task is to identify the material responsible for the Raman spectrum, then it is only necessary to compare the measured spectrum with the reference spectra of candidate materials. Since the Raman spectrum acts as a fingerprint of the scattering material, once a match is found, the unknown substance can be identified.

Often the identity of the sample is known, and Raman spectroscopy is being used to learn more about the material, e.g., the strength of its atomic bonds or the presence of elastic strains in a crystalline matrix. Such information is present in the Raman spectrum but it must now be analyzed carefully in order to characterize the structure of the material.

Once a vibrational Raman spectrum is obtained, the first task is to identify the vibrational modes that contribute to each peak in the spectrum. Group theoretical techniques enormously simplify the analyses of vibrational spectra. At this point, the rules of group theory that are used in vibrational spectroscopy will be cited and examples of their use will then be given. In doing so, it is worth recalling the warning issued by David Bishop in the preface to his textbook on Group Theory (Bishop, 1973, p. vii): "The mathematics involved in actually applying, as opposed to deriving group theoretical formulae is quite trivial. It involves little more than adding and multiplying. It is in fact possible to make the applications, by filling in the necessary formulae in a routine way, without even understanding where the formulae have come from. I do not, however, advocate this practice."

Although the approach taken in the body of this unit ignores Bishop's advice, the intent is to demonstrate the ease with which vibrational spectra can be analyzed by using the tools of group theory. One researcher used the automobile analogy to illustrate the usefulness of this approach to group theory. Driving an automobile can be a very useful practice. Doing so can be accomplished without any knowledge of the internal combustion engine. Hopefully, the elegance of the methodology, which should be apparent from reading this appendix, will encourage the reader to acquire a firm understanding of the principles of group theory by consulting any number of texts that provide an excellent introduction to the subject (see Literature Cited, e.g., Hammermesh, 1962; Koster, 1957; Leech and Newman, 1969; Mariot, 1962; Meijer and Bauer, 1962; Weyl, 1931).

## Group Theoretical Tools for Analyzing Vibrational Spectra

**Point Groups and Matrix Representations of Symmetry Operations.** The collection of symmetry operations of a molecule make up its point group. A point group is a collection of symmetry operations that are linked to one another by the following rules: (1) The identity operation is a member of the set. (2) The operations multiply associatively. (3) If $R$ and $S$ are elements, then so is $RS$. (4) The inverse of each element is also a member of the set. In satisfying these requirements, the point groups meet the criteria for a mathematical group so that molecular symmetry can be investigated with the aid of group theory.

At this juncture, point groups, which describe the symmetry of molecules, and the use of group theoretical techniques to identify the selection rules for Raman active, molecular vibrations will be described. Although the symmetry of crystals is described by space groups, it will be demonstrated below that the Raman active vibrational modes of a crystal can be identified with the aid of the point group associated with the crystal space group. That is, the Raman spectrum of a crystal is equated to the Raman spectrum of a single unit cell, which is treated like a molecule. Specifically, the symmetry of the unit cell is described by one of 32 point groups. All that is presented about point groups and molecular symmetry will be directly applicable to the use of symmetry and group theory for analyses of vibrational spectra of solids. After completing the discussion of point groups and molecular symmetry, the link between point groups and vibrational spectroscopy of solids will be established.

There are a total of fourteen different types of point groups, seven with one principal axis of rotation; groups within these seven types form a series in which one group can be formed from one other group by the addition of a symmetry operation. The seven other types of point groups do not constitute a series and involve multiple axes of higher order. The symmetries of the vast number of different molecules can be completely represented by a small number of point groups.

The water molecule belongs to the point group $C_{2v}$, which consists of the symmetry elements $\sigma_{v(xz)}, \sigma_v(yz)$, and $C_2$, as illustrated in Figure 6, plus the identity operation $E$. Each symmetry operation can be represented by a matrix that describes how the symmetry element acts on the molecule. First, a basis function that represents the molecule is identified. Examples of basis functions for the water molecule are: (1) the 1$s$ orbitals of the two hydrogen atoms and the 2$s$ orbital of the oxygen atom; (2) the displacement coordinates of the molecule during normal-mode vibrations; and (3) the 1$s$ orbitals of the hydrogen atoms and the $2p_{x,y,z}$ orbitals of oxygen. In fact, there are any number of possible basis functions, although some are more efficient and effective than others in representing the molecule. The number of functions in the basis defines the dimensions of the matrix representations of the symmetry operations. In the case of a basis consisting of three components, e.g., the two 1$s$ orbitals of hydrogen atoms and the 2$s$ orbital of oxygen, each matrix would be $3 \times 3$. The action of the symmetry operation on the molecule would then be provided by an equation of the form

$$[f_1 f_2 f_3] = \begin{pmatrix} d_{11} & d_{12} & d_{13} \\ d_{21} & d_{22} & d_{23} \\ d_{31} & d_{32} & d_{33} \end{pmatrix} \begin{bmatrix} 2s \\ 2s_{\mathrm{a}} \\ 1s_{\mathrm{b}} \end{bmatrix} \qquad (31)$$

where $[f_1, f_2, f_3]$ represents the basis function following the action of the symmetry operation and $1s_{\mathrm{a}}$ represents the 1$s$ orbital on the "a" hydrogen atom. If the symmetry operation under consideration is the twofold axis of rotation,

$$D(R) = \begin{bmatrix} a & b & o \\ c & d & o \\ o & o & e \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} + [e]$$

**Figure 14.** Matrix representation of the symmetry operation $R$ is given as $D(R)$. In the form depicted, $D(R)$ can be visualized as the "direct sum" of a $2 \times 2$ and a $1 \times 1$ matrix.

then $[f_1, f_2, f_3]$ is $[2s, 1s_b, 1s_a]$ and the matrix representation of the twofold axis of rotation for this basis function is

$$D(C_2) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \tag{32}$$

In instances in which one component of the basis function is unchanged by all of the symmetry operations, each matrix representation has the form depicted in Figure 14. The matrix can be thought of as consisting of the sum of two matrices, a $1 \times 1$ matrix and a $2 \times 2$ matrix. This type of sum is not an arithmetic operation and is referred to as a "direct sum." The process of converting all of the matrices that represent each of the symmetry operations for a given basis to the direct sum of matrices of smaller dimensions is referred to as "reducing the representation." In some cases each $2 \times 2$ matrix representation of the symmetry operations is also diagonalized, and the representation can be reduced to the direct sum of two $1 \times 1$ matrices. When the matrix representations have been converted to the direct sum of matrices with the smallest possible dimensions, the process of reducing the representation is complete.

Obviously, there are an infinite number of matrix representations that could be developed for each point group, each arising from a different basis. Fortunately, and perhaps somewhat surprisingly, most of the information about the symmetry of a molecule is contained in the characters of the matrix representations. Knowledge of the full matrix is not required in order to use the symmetry of the molecule to help solve complex problems in quantum mechanics. Often, only the characters of the matrices representing the group symmetry operations are needed.

The many different matrix representations of the symmetry operations can be reduced via similarity transformations to block diagonalized form. In block diagonalized form, each matrix is seen to consist of the direct sum of a number of irreducible matrices. An irreducible matrix is one for which there is no similarity transformation that is capable of putting it into block diagonalized form. A reduced representation consists of a set of matrices, each of which represents one symmetry operation of the point group and each of which is in the identical block diagonalized form.

Now it turns out that the character of a matrix representation is the most important piece of information concerning the symmetry of the molecule, as mentioned above. The list composed of the characters of the matrix representations of the symmetry operations of the group is referred to as the symmetry species of the representation. For each point group there is a small number of unique symmetry species that are composed of the characters of irreducible matrices representing each of the symmetry operations. These special symmetry species are called irreducible representations of the point group. For each point group the number of irreducible representations is equal to the number of symmetry classes.

Remarkably, of the unlimited number of representations of a point group, corresponding to the unlimited number of bases that can be employed, the symmetry species of each representation can be expressed by a linear combination of symmetry species of irreducible representations. The list of the symmetry species of each irreducible representation therefore concisely indicates what is unique about the symmetry of the particular point group. This information is provided in character tables of the point groups.

**Character Tables.** The character table for the point group $C_{2v}$ is presented in Table 1. The top row lists the symmetry elements in the point group. The second through the fourth rows list the point group irreducible representations. The first column is the symbol used to denote the irreducible representation, e.g., $A_1$. The numbers in the character tables are the characters of the irreducible matrix representations of the symmetry operations for the various irreducible symmetry species. The right-hand column lists basis functions for the irreducible symmetry species. For example, if $x$ is used as a basis function for the point group $C_{2v}$, the matrix representations of the various symmetry operations will all be $1 \times 1$ matrices with character of 1. The resultant symmetry species, therefore, is 1, 1, 1, 1 and is labeled $A_1$. On the other hand, if $xy$ is the basis function, the matrix representations of all of the symmetry operations will consist of $1 \times 1$ matrices. The resultant symmetry species is 1, 1, $-1$, $-1$ and is labeled $A_2$. If a function is said to have the same symmetry as $x$, then that function will also serve as a basis for the irreducible representation $B_1$ in the point group $C_{2v}$. There are many possible basis functions for each irreducible representation. Those listed in the character tables have special significance for the analyses of rotational, vibrational, and electronic spectra. This will be made clear below. When a function serves as a basis of a representation, it is said to span that representation.

**Vibrational Selection Rules.** Where group theory is especially helpful is in deciding whether or not a particular integral, taken over a symmetric range, has a nonzero value. It turns out that if the integrand can serve as a basis for the $A_1$, totally symmetric irreducible representation, then the integral may have a nonzero value. If the integrand does not serve as a basis for the $A_1$ irreducible representation, then the integral will necessarily be zero.

The symmetry species that is spanned by the products of two functions is obtained by forming the "direct product" of the symmetry species spanned by each function. Tables are available that list the direct products of all possible combinations of irreducible representations for each point group (see, e.g., Wilson et al., 1955; Bishop, 1973; Atkins, 1984; Kettle, 1987).

Thus, if we want to know whether or not a Raman transition from the ground-state vibrational mode $n$ to the first vibrationally excited state $m$ can occur, we need to examine the six integrals of the form

$$\int \psi_m^* \alpha_{ij} \psi_n dV \tag{33}$$

one for each of the six independent values of $\alpha_{ij}$ (or fewer depending on the symmetry of the molecule or crystal). If any one is nonzero, the Raman transition may occur. If all are zero, the Raman transition will be symmetry forbidden.

The Raman selection rule states that a Raman transition between two vibrational states $n$ and $m$ is allowed if the product $\psi_m^* \psi_n$ of the two wave functions describing the $m$ and $n$ vibrational states has the same symmetry species as at least one of the six components of $\alpha_{ij}$. This rule reflects the fact that for the integral $\int f_1 f_2 f_3 \, dV$ to have a nonzero value when taken over the symmetric range of variables $V$, it is necessary that the integrand, which is the product of the three functions $f_1$, $f_2$, and $f_3$, must span a symmetry species that is or contains the totally symmetric irreducible representation of the point group.

The symmetry species spanned by the product of three functions is obtained from the direct product of the symmetry species of each function (see, e.g., Wilson et al., 1955; Bishop, 1973; Atkins, 1984; Kettle, 1987). For the symmetry species of a direct product to contain the totally symmetric irreducible representation, it is necessary that the two functions span the same symmetry species or have a common irreducible representation in their symmetry species. The function $x^2$ acts as a basis for the $A_1$ irreducible representation of the point group $C_{2v}$. A basis for the irreducible representation $B_1$ is given as $x$. The function $x^2x$ would not span the irreducible representation $A_1$, and so its integral over all space would be zero. The function $xz$ is also a basis for the irreducible representation $B_1$ in the point group $C_{2v}$. Hence, the function $xxz(=x^2z)$ spans the irreducible representation $A_1$ and its integral over all space could be other than zero. Thus, for the integrand to span the totally symmetric irreducible representation of the molecular point group, it is necessary that the representation spanned by the product $\psi_m^* \psi_n$ be the same as the representation spanned by $\alpha_{ij}$. This last statement is extremely useful for analyzing the probability of a Stokes Raman transition from the vibrational ground state $n$ to the first vibrational excited state $m$ or for an anti-Stokes Raman transition from the first vibrational excited state $m$ to the vibrational ground state $n$. The vibrational ground state spans the totally symmetric species. Consequently, the direct product $\psi_m^* \psi_n$ spans the symmetry species of the first vibrational excited state $m$. Hence, the Raman selection rule means that the symmetry species of $\alpha_{ij}$ must be identical to the symmetry species of the first vibrational excited state.

The symmetry of $\alpha_{ij}$ can be obtained by showing that it transforms under the action of a symmetry operation of the point group in the same way as a function of known symmetry. Here, $\alpha_{ij}$ relates $p_i$ to $\alpha_{ij}$. If a symmetry operation $R$ is applied to the molecule/crystal, then the applied electric field is transformed to $R = \mathbf{D}(R)$, where $\mathbf{D}(R)$ is the matrix representation of $R$. The polarizability, which expresses the proportionality between the applied electric field and the induced polarization of the molecule/crystal, must transform in a manner related to but in general different from that of the applied electric field because the direction of $\mathbf{p}$ will generally be different from the direction of $\varepsilon_j$. The polarization of the crystal induced by the applied electric field is transformed to $Rp = D(R)p$ as a result of the action of the symmetry operation $R$. Consequently, the polarizability transforms as $RR\alpha = D(R)D(R)\alpha$. The transformation of $\alpha$ can be derived as follows (taken from Bishop, 1973).

If the symmetry operation $R$ transforms the coordinate system from $x$ to $x'$, then (summations are over all repeated indices)

$$\begin{aligned} p_i(x') &= \sum \alpha_{ij}(x')\varepsilon_j(x') \\ &= \sum \alpha_{ij}(x') \sum D_{jk}(R)\varepsilon_k(x) \end{aligned} \tag{34}$$

$$p_l(x) = \sum D_{ml}(R) p_m(x') \tag{35}$$

$$p_l(x) = \sum D_{ml}(R) \sum \alpha_{im}(x') \sum D_{jk}(R) \; {}_k(x) \tag{36}$$

Substituting into the last equation the expression

$$p_l(x) = \sum \alpha_{lk}(x) \; {}_k(x) \tag{37}$$

gives

$$\alpha_{lk}(x) = \sum \sum D_{ml}(R) D_{jk}(R) \alpha_{mj}(x') \tag{38}$$

This describes the action of the symmetry operation $R$ on the polarizability $\alpha$.

Now,

$$x_l = \sum D_{ml}(R) x_m \tag{39}$$

and

$$x_k = \sum D_{jk}(R) x_j \tag{40}$$

so

$$x_l x_k = \sum \sum D_{ml}(R) D_{jk}(R) x_m x_j \tag{41}$$

Comparison of Equations 19 and 22 indicates that $\alpha_{lk}$ transforms as the function $x_l x_k$. Any symmetry species spanned by the function $x_m x_j$ will also be spanned by $\alpha_{mj}$. Thus, any vibrational mode that spans the same symmetry species as $x_l x_k$ will be Raman active. This information is included in the character tables of the point groups. In the column that lists the basis functions for the various symmetry species of the point group are listed quadratic functions such as $x^2, yz, x^2 + y^2, x^2 + y^2 - 2z^2, \ldots$.

By way of illustration, the character table indicates that if a molecule with point group symmetry $C_{2v}$ has a vibrational mode that spans the symmetry species $A_1$, it will be Raman active since this symmetry species is also spanned

by the quadratic function $x^2$. Any vibrational mode that spans a symmetry species that has at least one quadratic function (e.g., $x^2$, $yz$) as a basis will be Raman active.

### Raman Active Vibrational Modes of Solids

It is now possible to illustrate how Raman spectroscopy can identify the symmetry species of the vibrational modes of a single crystal. To begin, it is important to recognize that the symmetry of a molecule may be lowered when it is part of a crystal. This can be appreciated by considering a simpler case of symmetry reduction: the change in symmetry of a sulfate anion as a consequence of its adsorption on a solid surface. As illustrated in Figure 15, the sulfate anion consists of oxygen nuclei located on the four corners of a tetrahedron and the sulfur nucleus positioned at the geometric center of the tetrahedron. The anion exhibits $T_d$ point group symmetry. If the sulfate is adsorbed on the surface of a solid through a bond formed between the solid and one of the oxygen anions of the sulfate, then the symmetry of the sulfate is lowered to $C_{3v}$. If two oxygen-surface bonds are formed, in either a bridging or nonbridging configuration, the symmetry of the sulfate is further lowered to $C_{2v}$. When present in a crystal, a molecule or anion will exhibit either the same or lower symmetry than the free molecule or anion.

The distinctive feature of the symmetry of a crystal is the presence of translational symmetry elements. In fact, it is possible to generate the entire lattice of a crystal by the repeated operation of the three basic primitive translation vectors that define the unit cell. When all possible arrays of lattice points in three dimensions are considered, it turns out that there are only fourteen distinctive types of lattices, called Bravais lattices. The Bravais lattice may not completely describe the symmetry of a crystal. The unit cell may have an internal structure that is not completely

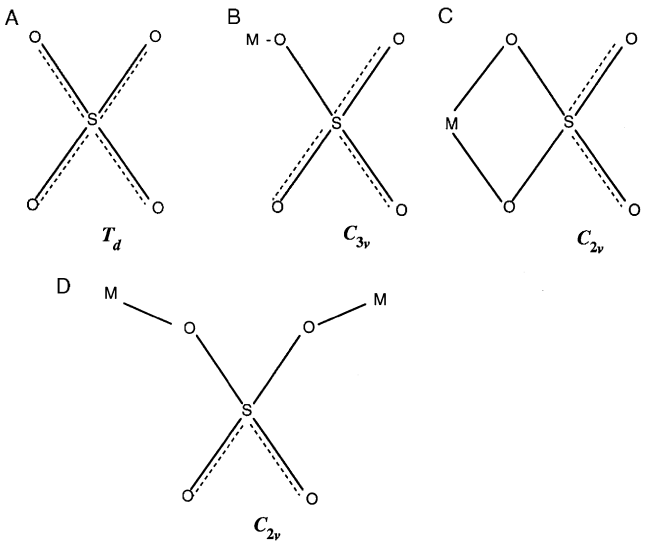



**Figure 15.** Influence of bonding on the symmetry of sulfate: (**A**) free sulfate; (**B**) unidentate sulfate; (**C**) bidentate sulfate; (**D**) bidentate sulfate in the bridging configuration.

specified by the symmetry elements of the bare lattice. The symmetry of the crystal is completely specified by its space group, which is a mathematical group that contains translational operations and that may contain as many as three additional types of symmetry elements: rotations (proper and improper), which are the point group elements, screw axes, and glide planes. The latter two elements may be conceptually subdivided into a rotation (proper = screw axis; improper = glide plane) plus a rational fraction of a primitive lattice translation.

Lattices can be brought into themselves by the operations of certain point groups. Each Bravais lattice type is compatible with only a particular set of point groups, which are 32 in number and are referred to as crystallographic point groups. The appropriate combinations of the 14 Bravais lattices and the 32 crystallographic point groups result in the 230 three-dimensional space groups.

The restrictions that a lattice imposes on the rotational symmetry elements of the space group can be readily illustrated. Each lattice point is located at the tip of a primitive lattice vector given by

$$\mathbf{T} = n_1\mathbf{t}_1 + n_2\mathbf{t}_2 + n_3\mathbf{t}_3 \tag{42}$$

where the $\mathbf{t}_i$ are the basic primitive translation vectors and the $n_i$ are positive and negative integers including zero. The rotation of the lattice translates the points to new locations given by

$$\mathbf{T}' = \boldsymbol{R}T = R_{ij}n_j\mathbf{t}_j \tag{43}$$

where $R_{ij}n_j$ must be integers for all values of $n_j$, which are integers. Hence, the $R_{ij}$ must be integers as well. For a proper or improper rotation of $\theta$ about the $z$ axis,

$$\boldsymbol{R} = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & \pm 1 \end{bmatrix} \tag{44}$$

The requirement of integral values for each element $R_{ij}$ means, in particular, that

$$\mathrm{Tr}\ \boldsymbol{R} = 2\cos\theta \pm 1 = \text{integer} \tag{45}$$

The plus sign corresponds to a proper rotation, while the minus sign corresponds to an improper rotation (e.g., reflection or inversion). Thus, the requirement of integral values of $R_{ij}$ limits the possible rotation angles $\theta$ to

$$\cos\theta = (n - \pm 1)/2 \Rightarrow \theta = 0, \pi/3, \pi/2, 2\pi/3, 2\pi \tag{46}$$

Crystallographic point groups may therefore contain the following proper rotations: $C_1$, $C_2$, $C_3$, $C_4$, and $C_6$, where $C_n$ indicates a rotation angle of $180^\circ/n$ about an axis of the crystal.

Within each space group is an invariant subgroup consisting of the primitive translational operations. An invariant group is one in which a conjugate transformation

produces another member of the group; i.e., if for all elements $X$ and $t$ of the group $G$, $X^{-1}tX = t'$, where $t$ and $t'$ are elements of the same subgroup of $G$, then this subgroup is an invariant subgroup of $G$.

A translational operation is represented by the symbol $\underline{\varepsilon}|\mathbf{t}$, which is a particular case of the more general symbol for a rotation of $\alpha$ followed by a translation of $\mathbf{a}$, $\underline{\alpha}|\mathbf{a}$. The notation $\underline{\varepsilon}$ represents a rotation of $0^\circ$. The invariance of the translational subgroup is demonstrated by

$$\begin{aligned}
(\underline{\alpha}|\mathbf{a})\mathbf{x} &= \underline{\alpha}\mathbf{x} + \mathbf{a} \\
(\underline{\beta}|\mathbf{b})(\underline{\alpha}|\mathbf{a})\mathbf{x} &= \underline{\beta\alpha}\mathbf{x} + \underline{\beta}\mathbf{a} + \mathbf{b} \\
&= (\underline{\beta\alpha}|\underline{\beta}\mathbf{a} + \mathbf{b})\mathbf{x}
\end{aligned}$$

Since $(\underline{\alpha}|\mathbf{a})^{-1}(\underline{\alpha}|\mathbf{a}) = (\underline{\varepsilon}|0)$

$(\underline{\alpha}|\mathbf{a})^{-1}$ must be given by $(\underline{\alpha}^{-1}| - \underline{\alpha}^{-1}\mathbf{a})$

$$\begin{aligned}
(\underline{\alpha}|\mathbf{a})^{-1}(\underline{\varepsilon}|\mathbf{t})(\underline{\alpha}|\mathbf{a})\mathbf{x} &= (\underline{\alpha}|\mathbf{a})^{-1}(\underline{\varepsilon}|\mathbf{t})(\underline{\alpha}\mathbf{x} + \mathbf{a}) \\
(\underline{\alpha}|\mathbf{a})^{-1}(\underline{\alpha}\mathbf{x} + \mathbf{t} + \mathbf{a}) &= \underline{\alpha}^{-1}\underline{\alpha}\mathbf{x} + \underline{\alpha}^{-1}\mathbf{t} + \underline{\alpha}^{-1}\mathbf{a} - \underline{\alpha}^{-1}\mathbf{a} \\
&= \mathbf{x} + \underline{\alpha}^{-1}\mathbf{t} \\
&= (\underline{\varepsilon}|\underline{\alpha}^{-1}\mathrm{t})\mathbf{x} \\
&= (\underline{\varepsilon}|\mathbf{t}')\mathbf{x} \qquad (47)
\end{aligned}$$

Thus, a conjugate transformation of a translation operator $\underline{\varepsilon}|\mathbf{t}$ using a general rotation plus translation operator $\underline{\alpha}|\mathbf{a}$ that is also a member of the group $G$ produces another simple translation operator $\underline{\varepsilon}|\mathbf{t}'$, demonstrating that the translational subgroup is invariant to the conjugacy operation.

The invariant property of the translational subgroup is exploited quite heavily in developing expressions for the irreducible representations of space groups. The difficulty in dealing with the symmetry of a crystal and in analyzing the vibrational modes of a crystal stems from the fact that the translational subgroup is infinite in size. Stated slightly differently, a crystal consists of a large number of atoms so that the number of normal vibrational modes is huge, i.e., $\approx 10^{23}$. Fortunately, group theory provides a method for reducing this problem to a manageable size.

Most textbooks dealing with applications of group theory to crystals develop cofactor expressions for space groups. The cofactor group is isomorphic with the crystallographic point group providing the space group is symmorphic. A symmorphic space group consists of rotational and translational symmetry elements. A nonsymmorphic space group contains at least one symmetry element $(\underline{\alpha}|\mathbf{a})$ in which the rotation $\alpha$, all by itself, and/or the translation $\mathbf{a}$, all by itself, are not symmetry elements of the space group. Of the 230 three-dimensional space groups, only 73 are symmorphic. Consequently, the use of factor groups is of no benefit for analyzing the vibrational modes of crystals in the 157 nonsymmorphic space groups.

Since the use of cofactor groups is so common, albeit of limited value, the approach will be summarized here and then a more general approach that addresses both symmorphic and nonsymmorphic space groups will be developed.

The translational factor group $G|T$ of the symmorphic space group $G$ consists of

$$G|T = T + \sum s_i T \qquad (i = 2, 3, \ldots, n) \qquad (48)$$

where the $s_i$ are the nontranslational symmetry elements except the identity element ($s_1 = E$) of the symmorphic space group $G$ whose crystallographic point group is of order $n$; $T$ is the translational symmetry elements of $G$. The translational factor group is distinct from the group $G$ because $T$ is treated as a single entity, rather than as the infinite number of lattice translational operations that it is in the group $G$. The factor group has a different multiplication rule than the space group and $T$ acts as the identity element. As long as $G$ is symmorphic, the factor group $G|T$ is obviously isomorphic with the crystallographic point group of $G$. Because of the isomorphic relationship of the two groups, the irreducible representations of the crystallographic point group will also serve as the irreducible representations of the factor group $G|T$. The irreducible representations of the factor group provide all the symmetry information about the crystal that is needed to analyze its vibrational motion. On the other hand, if $G$ is nonsymmorphic, then it contains at least one element $s_j$ of the form $[\underline{\alpha}|\mathbf{v}(\alpha)]$, where $\alpha$ and/or $\mathbf{v}(\alpha)$ are not group elements. As a consequence, $G|T$ is no longer symmorphic with any crystallographic point group.

The value of this approach, which makes use of cofactor groups, is that the total number of group elements have been reduced from $g \times 10^{23}$, where $g$ is the number of point group operations, to $g$. Furthermore, the factor group $G|T$ is isomorphic with the crystallographic point group. Consequently, the irreducible representations of the point group serve as irreducible representations of the factor group. How does this help in the analyses of vibrational modes? Recall that the only Raman active modes are characterized by $k = 0$ (i.e., approximately infinite wavelength) and nonzero frequency. The modes of $k = 0$ consist of corresponding atoms in each unit cell moving in phase. The number of normal modes of this type of motion is given by $3N$, where $N$ is the number of atoms in the unit cell. The symmetry of the unit cell is just that of the factor group. Thus, the irreducible representations of the crystallographic point group provide the irreducible representations of the Raman active vibrational modes of the crystal.

As mentioned above, a more general approach to the group theoretical analysis of crystal vibrational modes will be developed in place of the use of factor groups. This approach is enunciated by Lax (1974).

The group multiplication of symmetry elements of space groups is denoted by

$$\begin{aligned}
[\underline{\alpha}|\mathbf{v}(\alpha)][\underline{\beta}|\mathbf{v}(\beta)]\mathbf{r} &= [\underline{\alpha}|\mathbf{v}(\alpha)][\underline{\beta}\mathbf{r} + \mathbf{v}(\beta)] \\
&= \underline{\alpha\beta}\mathbf{r} + \underline{\alpha}\mathbf{v}(\beta) + \mathbf{v}(\alpha) \\
&= \underline{\alpha\beta}\mathbf{r} + \underline{\alpha}\mathbf{v}(\beta) + \mathbf{v}(\alpha) + \mathbf{v}(\alpha\beta) - \mathbf{v}(\alpha\beta) \\
&\quad \underline{\alpha\beta}\mathbf{r} + \mathbf{v}(\alpha\beta) + \underline{\alpha}\mathbf{v}(\beta) + \mathbf{v}(\alpha) - \mathbf{v}(\alpha\beta) \\
&= \alpha\beta\mathbf{r} + \mathbf{v}(\alpha\beta) + \mathbf{t} \\
\text{where} \quad \mathbf{t} &= \underline{\alpha}\mathbf{v}(\beta) + \mathbf{v}(\alpha) - \mathbf{v}(\alpha\beta) \\
&= (\underline{\varepsilon}|\mathbf{t})(\underline{\alpha\beta}|\mathbf{v}(\alpha\beta)) \qquad (49)
\end{aligned}$$